CHAPTER 2

STATIONARY TIME-SERIES MODELS

Learning Objectives

1. Describe the nature of stochastic linear difference equations.
2. Develop the tools used in estimating ARMA models.
3. Consider the time-series properties of stationary and nonstationary models.
4. Consider various test statistics to check for model adequacy. Several examples of estimated ARMA models are analyzed in detail. It is shown that how a properly estimated model can be used for forecasting.
5. Derive the theoretical autocorrelation function for various ARMA processes.
6. Derive the theoretical partial autocorrelation function for various ARMA processes.
7. Show how the Box–Jenkins methodology relies on the autocorrelations and partial autocorrelations in model selection.
8. Develop the complete set of tools for Box–Jenkins model selection.
9. Examine the properties of time-series forecasts.
10. Illustrate the Box–Jenkins methodology using a model of the term structure of interest rates.
11. Show how to model series containing seasonal factors.
12. Develop diagnostic testing for model adequacy.
13. Show that combined forecasts typically outperform forecasts from a single model.

1. STOCHASTIC DIFFERENCE EQUATION MODELS

In this chapter, we continue to work with **discrete**, rather than continuous, time-series models. Recall from the discussion in Chapter 1 that we can evaluate the function $y = f(t)$ at t_0 and $t_0 + h$ to form

$$\Delta y = f(t_0 + h) - f(t_0)$$

As a practical matter, most economic time-series data are collected for discrete time periods. Thus, we consider only the equidistant intervals t_0, $t_0 + h$, $t_0 + 2h$, $t_0 + 3h$, … and conveniently set $h = 1$. Be careful to recognize, however, that a discrete time series implies that t, but not necessarily y_t, is discrete. For example, although Scotland's annual rainfall is a continuous variable, the sequence of such annual rainfall totals for years 1 through t is a discrete time series. In many economic applications, t refers to "time" so that h represents the change in time. However, t need not refer to the type of time interval as measured by a clock or calendar. Instead of allowing our measurement units to be minutes, days, quarters, or years, t can refer to an ordered event number. We could let y_t denote the outcome of spin t on a roulette wheel; y_t can then take on any of the 38 values 00, 0, 1, … , 36.

A discrete variable y is said to be a **random** variable (i.e., stochastic) if, for any real number r, there exists a probability $p(y \leq r)$ that y will take on a value less than or equal to r. This definition is fairly general; in common usage, it is typically implied that there is at least one value of r for which $0 < p(y = r) < 1$. If there is some r for which $p(y = r) = 1$, y is deterministic rather than random.

It is useful to consider the elements of an observed time series $\{y_0, y_1, y_2, \ldots, y_t\}$ as being realizations (i.e., outcomes) of a stochastic process. As in Chapter 1, we continue to let the notation y_t to refer to an element of the entire sequence $\{y_t\}$. In our roulette example, y_t denotes the outcome of spin t on a roulette wheel. If we observe spins 1 through T, we can form the sequence $y_1, y_2, \ldots, y_T$ or, more compactly, $\{y_t\}$. In the same way, the term y_t could be used to denote gross domestic product (GDP) in time period t. Since we cannot forecast GDP perfectly, y_t is a random variable. Once we learn the value of GDP in period t, y_t becomes one of the realized values from a stochastic process. (Of course, measurement error may prevent us from ever knowing the "true" value of GDP.)

For discrete variables, the probability distribution of y_t is given by a formula (or table) that specifies each possible realized value of y_t and the probability associated with that realization. If the realizations are linked across time, there exists the joint probability distribution $p(y_1 = r_1, y_2 = r_2, \ldots, y_T = r_T)$ where r_i is the realized value of y in period i. Having observed the first t realizations, we can form the expected value of $y_{t+1}, y_{t+2}, \ldots$, conditioned on the observed values of y_1 through y_t. This conditional mean, or expected value, of y_{t+i} is denoted by $E_t[y_{t+i} | y_t, y_{t-1}, \ldots, y_1]$ or $E_t y_{t+i}$.

Of course, if y_t refers to the outcome of spinning a fair roulette wheel, the probability distribution is easily characterized. In contrast, we may never be able to completely describe the probability distribution for GDP. Nevertheless, the task of economic theorists is to develop models that capture the essence of the true data-generating process. Stochastic difference equations are one convenient way of modeling dynamic economic processes. To take a simple example, suppose that the Federal Reserve's money supply target grows 3% each year. Hence,

$$m_t^* = 1.03 m_{t-1}^* \tag{2.1}$$

so that, given the initial condition m_0^*, the particular solution is

$$m_t^* = (1.03)^t m_0^*$$

where m_t^* = the money supply target in year t;

m_0^* = the initial condition for the target money supply in period zero.

Of course, the actual money supply (m_t) and the target need not be equal. Suppose that, at the end of period $t-1$, there exists m_{t-1} outstanding dollars that are carried forward into period t. Hence, at the beginning of t, there are m_{t-1} dollars so that the gap between the target and the actual money supply is $m_t^* - m_{t-1}$. Suppose that the Fed cannot perfectly control the money supply but attempts to change the money supply by ρ percentage ($\rho < 100\%$) of any gap between the desired and actual money supply. We can model this behavior as

$$\Delta m_t = \rho[m_t^* - m_{t-1}] + \varepsilon_t$$

or using (2.1), we obtain

$$m_t = \rho(1.03)^t m_0^* + (1-\rho)m_{t-1} + \varepsilon_t \tag{2.2}$$

where ε_t is the uncontrollable portion of the money supply.

We assume that the mean of ε_t is zero in all time periods.

Although the economic theory is overly simple, the model does illustrate the key points discussed earlier. Note the following:

1. Although the money supply is a continuous variable, (2.2) is a discrete difference equation. Since the forcing process $\{\varepsilon_t\}$ is stochastic, the money supply is stochastic; we can call (2.2) a linear stochastic difference equation.
2. If we knew the distribution of $\{\varepsilon_t\}$, we could calculate the distribution for each element in the $\{m_t\}$ sequence. Since (2.2) shows how the realizations of the $\{m_t\}$ sequence are linked across time, we would be able to calculate the various joint probabilities. Notice that the distribution of the money supply sequence is completely determined by the parameters of the difference equation (2.2) and the distribution of the $\{\varepsilon_t\}$ sequence.
3. Having observed the first t observations in the $\{m_t\}$ sequence, we can make forecasts of $m_{t+1}, m_{t+2}, \ldots$. For example, updating (2.2) by one period and taking the conditional expectation, the forecast of m_{t+1} is $E_t m_{t+1} = \rho(1.03)^{t+1} m_0^* + (1-\rho)m_t$.

Before we proceed too far along these lines, let us go back to the basic building block of discrete stochastic time-series models: the **white-noise** process. A sequence $\{\varepsilon_t\}$ is a white-noise process if each value in the sequence has a mean of zero, a constant variance, and is uncorrelated with all other realizations. Formally, if the notation $E(x)$ denotes the theoretical mean value of x, the sequence $\{\varepsilon_t\}$ is a white-noise process if for each time period t

$$\begin{aligned}
E(\varepsilon_t) &= E(\varepsilon_{t-1}) = \cdots = 0 \\
E(\varepsilon_t^2) &= E(\varepsilon_{t-1}^2) = \cdots = \sigma^2 && [\text{or } \operatorname{var}(\varepsilon_t) = \operatorname{var}(\varepsilon_{t-1}) = \cdots = \sigma^2] \\
E(\varepsilon_t\, \varepsilon_{t-s}) &= E(\varepsilon_{t-j}\, \varepsilon_{t-j-s}) \\
&= 0 \text{ for all } j \text{ and } s && [\text{or } \operatorname{cov}(\varepsilon_t, \varepsilon_{t-s}) = \operatorname{cov}(\varepsilon_{t-j}, \varepsilon_{t-j-s}) = 0]
\end{aligned}$$

In the remainder of this text, $\{\varepsilon_t\}$ will always refer to a white-noise process and σ^2 will refer to the variance of that process. When it is necessary to refer to two or more white-noise processes, symbols such as $\{\varepsilon_{1t}\}$ and $\{\varepsilon_{2t}\}$ will be used. Now, use a white-noise process to construct the more interesting time series

$$x_t = \sum_{i=0}^{q} \beta_i \varepsilon_{t-i} \tag{2.3}$$

For each period t, x_t is constructed by taking the values $\varepsilon_t, \varepsilon_{t-1}, \ldots, \varepsilon_{t-q}$ and multiplying each by the associated value of β_i. A sequence formed in this manner is called a **moving average** of order q and is denoted by MA(q). To illustrate a typical moving average process, suppose you win \$1 if a fair coin shows a head and lose \$1 if it shows a tail. Denote the outcome on toss t by ε_t (i.e., for toss t, ε_t is either +\$1 or −\$1). If you want to keep track of your hot streaks, you might want to calculate your average winnings on the last four tosses. For each coin toss t, your average payoff on the last four tosses is $1/4\varepsilon_t + 1/4\varepsilon_{t-1} + 1/4\varepsilon_{t-2} + 1/4\varepsilon_{t-3}$. In terms of (2.3), this sequence is a moving average process such that $\beta_i = 0.25$ for $i \leq 3$ and zero otherwise.

Although the $\{\varepsilon_t\}$ sequence is a white-noise process, the constructed $\{x_t\}$ sequence will *not* be a white-noise process if two or more of the β_i differ from zero. To illustrate using an MA(1) process, set $\beta_0 = 1$, $\beta_1 = 0.5$, and all other $\beta_i = 0$. In this circumstance, $E(x_t) = E(\varepsilon_t + 0.5\varepsilon_{t-1}) = 0$ and $\text{var}(x_t) = \text{var}(\varepsilon_t + 0.5\varepsilon_{t-1}) = 1.25\sigma^2$. You can easily convince yourself that $E(x_t) = E(x_{t-s})$ and that $\text{var}(x_t) = \text{var}(x_{t-s})$ for all s. Hence, the first two conditions for $\{x_t\}$ to be a white-noise process are satisfied. However, $E(x_t x_{t-1}) = E[(\varepsilon_t + 0.5\varepsilon_{t-1})(\varepsilon_{t-1} + 0.5\varepsilon_{t-2})] = E(\varepsilon_t\varepsilon_{t-1} + 0.5(\varepsilon_{t-1})^2 + 0.5\varepsilon_t\varepsilon_{t-2} + 0.25\varepsilon_{t-1}\varepsilon_{t-2}) = 0.5\sigma^2$. Given that there exists a value of $s \neq 0$ such that $E(x_t x_{t-s}) \neq 0$, the $\{x_t\}$ sequence is not a white-noise process.

Exercise 1 at the end of this chapter asks you to find the mean, variance, and covariance of your hot streaks in coin tossing. For practice, you should work that exercise before continuing. If you are a bit "rusty" on the algebra of finding means, variances and covariances, you should also work through Exercises 2 and 3 and consult Section 2.3 of the *Supplementary Manual* to this text.

2. ARMA MODELS

It is possible to combine a moving average process with a linear difference equation to obtain an autoregressive moving average (ARMA) model. Consider the pth order difference equation

$$y_t = a_0 + \sum_{i=1}^{p} a_i y_{t-i} + x_t \tag{2.4}$$

Now let $\{x_t\}$ be the MA(q) process given by (2.3) so that we can write

$$y_t = a_0 + \sum_{i=1}^{p} a_i y_{t-i} + \sum_{i=0}^{q} \beta_i \varepsilon_{t-i} \tag{2.5}$$

We follow the convention of normalizing units so that β_0 is always equal to unity. If the characteristic roots of (2.5) are all in the unit circle, $\{y_t\}$ is called an ARMA model for y_t. The autoregressive part of the model is the *difference equation* given by the homogeneous portion of (2.4) and the moving average part is the $\{x_t\}$ sequence. If the homogeneous part of the difference equation contains p lags and the model for

x_t contains q lags, the model is called an ARMA(p, q) model. If $q = 0$, the process is called a pure autoregressive process denoted by AR(p), and if $p = 0$, the process is a pure moving average process denoted by MA(q). In an ARMA model, it is perfectly permissible to allow p and/or q to be infinite. In this chapter, we consider only models in which all of the characteristic roots of (2.5) are within the unit circle. However, if one or more characteristic roots of (2.5) is greater than or equal to unity, the $\{y_t\}$ sequence is said to be an **integrated** process and (2.5) is called an autoregressive integrated moving average (ARIMA) model.

Treating (2.5) as a difference equation suggests that we can "solve" for y_t in terms of the $\{\varepsilon_t\}$ sequence. The solution of an ARMA(p, q) model expressing y_t in terms of the $\{\varepsilon_t\}$ sequence is the **moving average representation** of y_t. The procedure is no different from that discussed in Chapter 1. For the AR(1) model $y_t = a_0 + a_1 y_{t-1} + \varepsilon_t$, the moving average representation was shown to be

$$y_t = a_0/(1 - a_1) + \sum_{i=0}^{\infty} a_1^i \varepsilon_{t-i}$$

For the general ARMA(p, q) model, rewrite (2.5) using lag operators so that

$$\left(1 - \sum_{i=1}^{p} a_i L^i\right) y_t = a_0 + \sum_{i=0}^{q} \beta_i \varepsilon_{t-i}$$

so that the *particular* solution for y_t is

$$y_t = \left(a_0 + \sum_{i=0}^{q} \beta_i \varepsilon_{t-i}\right) \bigg/ \left(1 - \sum_{i=1}^{p} a_i L^i\right) \tag{2.6}$$

Fortunately, it will not be necessary for us to expand (2.6) to obtain the specific coefficient for each element in $\{\varepsilon_t\}$. The important point to recognize is that the expansion will yield an MA(∞) process. The issue is whether such an expansion is convergent so that the stochastic difference equation given by (2.6) is stable. As you will see in Section 3, the stability condition is that the roots of the polynomial $\left(1 - \Sigma a_i L^i\right)$ must lie outside the unit circle. It is also shown that *if y_t is a linear stochastic difference equation, the stability condition is a necessary condition for the time series $\{y_t\}$ to be stationary*.

3. STATIONARITY

Suppose that the quality control division of a manufacturing firm samples four machines each hour. Every hour, quality control finds the mean of the machines' output levels. The plot of each machine's hourly output is shown in Figure 2.1. If y_{it} represents machine y_i's output at hour t, the means ($\overline{y}_t$) are readily calculated as

$$\overline{y}_t = \sum_{i=1}^{4} y_{it}/4$$

For hours 5, 10, and 15, these mean values are 4.61, 5.14, and 5.03, respectively.

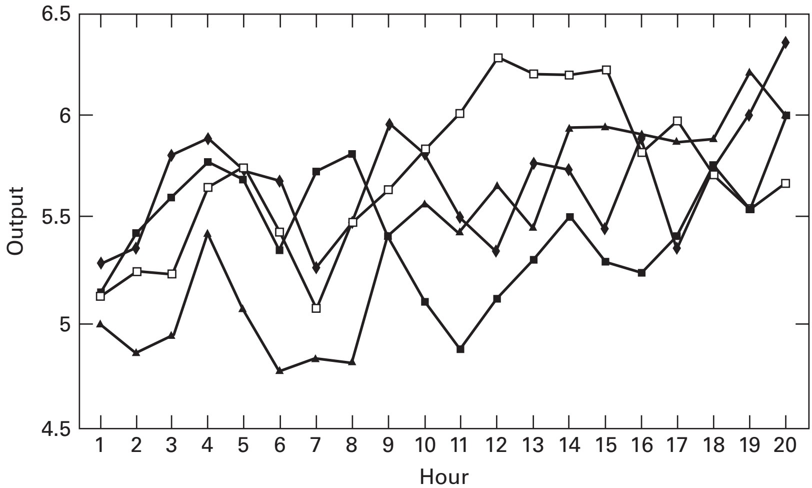


FIGURE 2.1 Hourly Output of Four Machines

The sample variance for each hour can similarly be constructed. Unfortunately, applied econometricians do not usually have the luxury of being able to obtain an **ensemble** (i.e., multiple time-series data of the same process over the same time period). Typically, we observe only one set of realizations for any particular series. Fortunately, if $\{y_t\}$ is a **stationary** series, the mean, variance, and autocorrelations can usually be well approximated by sufficiently long **time averages** based on the single set of realizations. Suppose that you observed only the output of machine 1 for 20 periods. If you knew that the output was stationary, you could approximate the mean level of output by

$$\bar{y}_t \cong \sum_{t=1}^{20} y_{1t}/20$$

In using this approximation, you would be assuming that the mean was the same for each period. Formally, a stochastic process having a finite mean and variance is **covariance stationary** if for all t and $t - s$,

$$E(y_t) = E(y_{t-s}) = \mu \tag{2.7}$$

$$E[(y_t - \mu)^2] = E[(y_{t-s} - \mu)^2] = \sigma_y^2 \qquad [\text{var}(y_t) = \text{var}(y_{t-s}) = \sigma_y^2] \tag{2.8}$$

$$E[(y_t - \mu)(y_{t-s} - \mu)] = E[(y_{t-j} - \mu)(y_{t-j-s} - \mu)] = \gamma_s$$
$$[\text{cov}(y_t, y_{t-s}) = \text{cov}(y_{t-j}, y_{t-j-s}) = \gamma_s] \tag{2.9}$$

where μ, σ_y^2, and γ_s are all constants.

In (2.9), allowing $s = 0$ means that γ_0 is equivalent to the variance of y_t. Simply put, a time series is covariance stationary if its mean and all autocovariances are unaffected by a change of time origin. In the literature, a covariance-stationary process is also referred to as a **weakly stationary**, second-order stationary, or wide-sense stationary process. (Note that a **strongly** stationary process need not have a finite mean and/or variance.) The text considers only covariance-stationary series so that there is no ambiguity in using the terms stationary and covariance stationary interchangeably. One further word about terminology: In multivariate models, the term autocovariance is reserved for the covariance between y_t and its own lags. Cross-covariance refers to the covariance between one series and another. In univariate time-series models, there is no ambiguity, and the terms autocovariance and covariance are used interchangeably.

For a covariance-stationary series, we can define the **autocorrelation** between y_t and y_{t-s} as

$$\rho_s \equiv \gamma_s / \gamma_0$$

where γ_0 and γ_s are defined by (2.9).

Since γ_s and γ_0 are time independent, the autocorrelation coefficients ρ_s are also time independent. Although the autocorrelation between y_t and y_{t-1} can differ from the autocorrelation between y_t and y_{t-2}, the autocorrelation between y_t and y_{t-1} must be identical to that between y_{t-s} and y_{t-s-1}. Obviously, $\rho_0 = 1$.

Stationarity Restrictions for an AR(1) Process

For expositional convenience, consider the necessary and sufficient conditions for an AR(1) process to be stationary. Let

$$y_t = a_0 + a_1 y_{t-1} + \varepsilon_t$$

where ε_t = white noise.

Suppose that the process started in period zero, so that y_0 is a deterministic initial condition. In Section 3 of Chapter 1, it was shown that the solution to this equation is (see also Question 4 at the end of this chapter)

$$y_t = a_0 \sum_{i=0}^{t-1} a_1^i + a_1^t y_0 + \sum_{i=0}^{t-1} a_1^i \varepsilon_{t-i} \tag{2.10}$$

Taking the expected value of (2.10), we obtain

$$Ey_t = a_0 \sum_{i=0}^{t-1} a_1^i + a_1^t y_0 \tag{2.11}$$

Updating by s periods yields

$$Ey_{t+s} = a_0 \sum_{i=0}^{t+s-1} a_1^i + a_1^{t+s} y_0 \tag{2.12}$$

Comparing (2.11) and (2.12), it is clear that both means are time dependent. Since Ey_t is not equal to Ey_{t+s}, the sequence cannot be stationary. However, if t is large, we can consider the limiting value of y_t in (2.10). If $|a_1| < 1$, the expression $(a_1)^t y_0$ converges to zero as t becomes infinitely large and the sum $a_0[1 + a_1 + (a_1)^2 + (a_1)^3 + \cdots]$ converges to $a_0/(1 - a_1)$. Thus, as $t \to \infty$ and if $|a_1| < 1$

$$\lim y_t = \frac{a_0}{1 - a_1} + \sum_{i=0}^{\infty} a_1^i \varepsilon_{t-i} \tag{2.13}$$

Now take expectation of (2.13) so that, for sufficiently large values of t, $Ey_t = a_0/(1 - a_1)$. Thus, the mean value of y_t is finite and time independent so that $Ey_t = Ey_{t-s} = a_0/(1 - a_1) \equiv \mu$ for all t. Turning to the variance, we find

$$\begin{aligned} E(y_t - \mu)^2 &= E[(\varepsilon_t + a_1\varepsilon_{t-1} + (a_1)^2\varepsilon_{t-2} + \cdots)^2] \\ &= \sigma^2[1 + (a_1)^2 + (a_1)^4 + \cdots] = \sigma^2/(1 - (a_1)^2) \end{aligned}$$

which is also finite and time independent. Finally, it is easily demonstrated that the limiting values of all autocovariances are finite and time independent:

$$\begin{aligned} E[(y_t - \mu)(y_{t-s} - \mu)] &= E\{[\varepsilon_t + a_1\varepsilon_{t-1} + (a_1)^2\varepsilon_{t-2} + \cdots] \\ &\qquad [\varepsilon_{t-s} + a_1\varepsilon_{t-s-1} + (a_1)^2\varepsilon_{t-s-2} + \cdots]\} \\ &= \sigma^2(a_1)^s[1 + (a_1)^2 + (a_1)^4 + \cdots] \\ &= \sigma^2(a_1)^s/[1 - (a_1)^2] \end{aligned} \tag{2.14}$$

In summary, if we can use the limiting value of (2.10), the $\{y_t\}$ sequence will be stationary. For any given y_0 and $|a_1| < 1$, it follows that t must be sufficiently large. Thus, if a sample is generated by a process that has recently begun, the realizations may not be stationary. It is for this very reason that many econometricians assume that the data-generating process has been occurring for an infinitely long time. In practice, the researcher must be wary of any data generated from a "new" process. For example, $\{y_t\}$ could represent the daily change in the dollar/mark exchange rate beginning immediately after the demise of the Bretton Woods fixed exchange rate system. Such a series may not be stationary due to that fact there were deterministic initial conditions (exchange rate changes were essentially zero in the Bretton Woods era). The careful researcher wishing to use stationary series might consider excluding some of these earlier observations from the period of analysis.

Little would change were we not given the initial condition. Without the initial value y_0, the sum of the homogeneous and particular solutions for y_t is

$$y_t = a_0/(1 - a_1) + \sum_{i=0}^{\infty} a_1^i\varepsilon_{t-i} + A(a_1)^t \tag{2.15}$$

where A is an arbitrary constant.

If you take the expectation of (2.15), it is clear that the $\{y_t\}$ sequence cannot be stationary unless the expression $A(a_1)^t$ is equal to zero. Either the sequence must have started infinitely long ago (so that $a_1{}^t = 0$) or the arbitrary constant A must be zero. Recall that the arbitrary constant is interpreted as a deviation from long-run equilibrium. The stability conditions can be stated succinctly:

1. The homogeneous solution must be zero. Either the sequence must have started infinitely far in the past or the process must always be in equilibrium (so that the arbitrary constant is zero).
2. The characteristic root a_1 must be less than unity in absolute value.

These two conditions readily generalize to all ARMA(p, q) processes. We know that the homogeneous solution to (2.5) has the form

$$\sum_{i=1}^{p} A_i\alpha_i^t$$

or, if there are m repeated roots,

$$\alpha \sum_{i=1}^{m} A_i t^i + \sum_{i=m+1}^{p} A_i \alpha_i^t$$

where A_i are all arbitrary constants, α is the repeated root, and α_i are the distinct roots.

If any portion of the homogeneous equation is present, the mean, variance, and all covariances will be time dependent. Hence, for any ARMA(p, q) model, stationarity necessitates that the homogeneous solution be zero. Section 4 addresses the stationarity restrictions for the particular solution.

4. STATIONARITY RESTRICTIONS FOR AN ARMA (*p, q*) MODEL

As a prelude to the stationarity conditions for the general ARMA(p, q) model, consider the restrictions necessary to ensure that an ARMA(2, 1) model is stationary. Since the magnitude of the intercept term does not affect the stability (or stationarity) conditions, set $a_0 = 0$ and write

$$y_t = a_1 y_{t-1} + a_2 y_{t-2} + \varepsilon_t + \beta_1 \varepsilon_{t-1} \tag{2.16}$$

From the previous section, we know that the homogeneous solution must be zero. As such, it is only necessary to find the particular solution. Using the method of undetermined coefficients, we can write the challenge solution as

$$y_t = \sum_{i=0}^{\infty} c_i \varepsilon_{t-i} \tag{2.17}$$

For (2.17) to be a solution of (2.16), the various c_i must satisfy

$$\begin{aligned}
&c_0\varepsilon_t + c_1\varepsilon_{t-1} + c_2\varepsilon_{t-2} + c_3\varepsilon_{t-3} + \cdots \\
&\quad = a_1(c_0\varepsilon_{t-1} + c_1\varepsilon_{t-2} + c_2\varepsilon_{t-3} + c_3\varepsilon_{t-4} + \cdots) \\
&\qquad + a_2(c_0\varepsilon_{t-2} + c_1\varepsilon_{t-3} + c_2\varepsilon_{t-4} + c_3\varepsilon_{t-5} + \cdots) + \varepsilon_t + \beta_1\varepsilon_{t-1}
\end{aligned}$$

To match coefficients on the terms containing $\varepsilon_t, \varepsilon_{t-1}, \varepsilon_{t-2}, \ldots$, it is necessary to set

1. $c_0 = 1$

2. $c_1 = a_1 c_0 + \beta_1 \qquad \Rightarrow c_1 = a_1 + \beta_1$

3. $c_i = a_1 c_{i-1} + a_2 c_{i-2} \qquad$ for all $i \geq 2$

The key point is that, for $i \geq 2$, the coefficients satisfy the difference equation $c_i = a_1 c_{i-1} + a_2 c_{i-2}$. If the characteristic roots of (2.16) are within the unit circle, the $\{c_i\}$ must constitute a convergent sequence. For example, reconsider the case in which $a_1 = 1.6$ and $a_2 = -0.9$, and let $\beta_1 = 0.5$. Worksheet 2.1 shows that the coefficients satisfying (2.17) are 1, 2.1, 2.46, 2.046, 1.06, −0.146, and so on (also see Worksheet 1.2 of the previous chapter).

WORKSHEET 2.1

COEFFICIENTS OF THE ARMA(2, 1) PROCESS: $y_t = 1.6y_{t-1} - 0.9y_{t-2} + \varepsilon_t + 0.5\varepsilon_{t-1}$

If we use the method of undetermined coefficients, the c_i must satisfy

$$c_0 = 1$$
$$c_1 = 1.16 + 0.5 \qquad \text{hence } c_1 = 2.1$$
$$c_2 = 1.6c_{i-1} - 0.9c_{i-2} \qquad \text{for all } i = 2, 3, 4, \ldots$$

Notice that the coefficients follow a second-order difference equation with imaginary roots. If we use de Moivre's Theorem, the coefficients will satisfy

$$c_i = 0.949^i \cdot \beta_1 \cos(0.567i + \beta_2)$$

Imposing the initial conditions for c_0 and c_1 yields

$$1 = \beta_1 \cos(\beta_2) \quad \text{and} \quad 2.1 = 0.949\beta_1 \cos(0.567 + \beta_2)$$

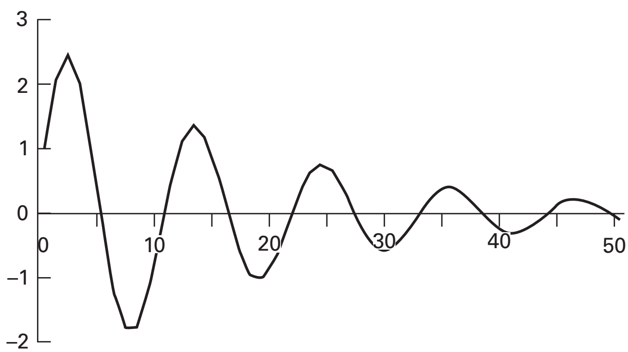
Since $\beta_1 = 1/\cos(\beta_2)$, we seek the solution to

$$\cos(\beta_2) - (0.949/2.1)\cos(0.567 + \beta_2) = 0$$

You can use a calculator or a trig table to verify that the solution for β_2 is -1.197 and the solution for β_1 is 2.739. Hence, the c_i must satisfy

$$(2.739) \cdot 0.949^i \cdot \cos(0.567i - 1.197)$$

Alternatively, we can use the initial values of c_0 and c_1 to find the other c_i by iteration. The sequence of the c_i is shown in the graph.



You can use a spreadsheet to verify that the values of c_0 through c_{10} are

i	0	1	2	3	4	5	6	7	8	9	10
c_i	1.00	2.10	2.46	2.046	1.06	−0.146	1.187	−1.786	−1.761	−1.226	−0.378

To verify that the $\{y_t\}$ sequence generated by (2.17) is stationary, take the expectation of (2.17) to form $Ey_t = Ey_{t-i} = 0$ for all t and i. Hence, the mean is finite and time invariant. Since the $\{\varepsilon_t\}$ sequence is assumed to be a white-noise process, the variance of y_t is constant and time independent; that is,

$$\begin{aligned}\text{var}(y_t) &= E[(c_0\varepsilon_t + c_1\varepsilon_{t-1} + c_2\varepsilon_{t-2} + c_3\varepsilon_{t-3} + \cdots)^2]\\ &= \sigma^2\sum_{i=0}^{\infty} c_i^2\end{aligned}$$

Hence, $\text{var}(y_t) = \text{var}(y_{t-s})$ for all t and s. Finally, the covariance between y_t and y_{t-s} is

$$\begin{aligned}\text{cov}(y_t, y_{t-1}) &= E[(\varepsilon_t + c_1\varepsilon_{t-1} + c_2\varepsilon_{t-2} + \cdots)(\varepsilon_{t-1} + c_1\varepsilon_{t-2} + c_2\varepsilon_{t-3} + c_3\varepsilon_{t-4} + \cdots)]\\ &= \sigma^2(c_1 + c_2c_1 + c_3c_2 + \cdots)\\ \text{cov}(y_t, y_{t-2}) &= E[(\varepsilon_t + c_1\varepsilon_{t-1} + c_2\varepsilon_{t-2} + \cdots)(\varepsilon_{t-2} + c_1\varepsilon_{t-3} + c_2\varepsilon_{t-4} + c_3\varepsilon_{t-5} + \cdots)]\\ &= \sigma^2(c_2 + c_3c_1 + c_4c_2 + \cdots)\end{aligned}$$

so that

$$\text{cov}(y_t, y_{t-s}) = \sigma^2(c_s + c_{s+1}c_1 + c_{s+2}c_2 + \cdots) \tag{2.18}$$

Thus, $\text{cov}(y_t, y_{t-s})$ is constant and independent of t. Conversely, if the characteristic roots of (2.16) do not lie within the unit circle, the $\{c_i\}$ sequence will *not* be convergent. As such, the $\{y_t\}$ sequence cannot be convergent.

It is not too difficult to generalize these results to the entire class of ARMA(p, q) models. Begin by considering the conditions ensuring the stationarity of a pure MA(∞) process. By appropriately restricting the β_i, all of the finite-order MA(q) processes can be obtained as special cases. Consider

$$x_t = \sum_{i=0}^{\infty} \beta_i\varepsilon_{t-i}$$

where $\{\varepsilon_t\}$ = a white-noise process with variance σ^2.

We have already determined that $\{x_t\}$ is not a white-noise process; now, the issue is whether $\{x_t\}$ is covariance stationary. Given conditions (2.7), (2.8), and (2.9), we ask the following:

1. Is the mean finite and time independent? Take the expected value of x_t and remember that the expectation of a sum is the sum of the individual expectations. Therefore,

$$\begin{aligned}E(x_t) &= E(\varepsilon_t + \beta_1\varepsilon_{t-1} + \beta_2\varepsilon_{t-2} + \cdots)\\ &= E\varepsilon_t + \beta_1E\varepsilon_{t-1} + \beta_2E\varepsilon_{t-2} + \cdots = 0\end{aligned}$$

Repeat the procedure with x_{t-s}:

$$E(x_{t-s}) = E(\varepsilon_{t-s} + \beta_1\varepsilon_{t-s-1} + \beta_2\varepsilon_{t-s-2} + \cdots) = 0$$

Hence, all elements in the $\{x_t\}$ sequence have the same finite mean ($\mu = 0$).

2. Is the variance finite and time independent? Form $\text{var}(x_t)$ as

$$\text{var}(x_t) = E[(\varepsilon_t + \beta_1\varepsilon_{t-1} + \beta_2\varepsilon_{t-2} + \cdots)^2]$$

Square the term in parentheses and take expectations. Since $\{\varepsilon_t\}$ is a white-noise process, all terms $E\varepsilon_t\varepsilon_{t-s} = 0$ for $s \neq 0$. Hence,

$$\begin{aligned}\text{var}(x_t) &= E(\varepsilon_t)^2 + (\beta_1)^2E(\varepsilon_{t-1})^2 + (\beta_2)^2E(\varepsilon_{t-2})^2 + \cdots \\ &= \sigma^2[1 + (\beta_1)^2 + (\beta_2)^2 + \cdots]\end{aligned}$$

As long as $\sum(\beta_i)^2$ is finite, it follows that $\text{var}(x_t)$ is finite. Thus, $\sum(\beta_i)^2$ being finite is a necessary condition for $\{x_t\}$ to be stationary. To determine whether $\text{var}(x_t) = \text{var}(x_{t-s})$, form

$$\begin{aligned}\text{var}(x_{t-s}) &= E[(\varepsilon_{t-s} + \beta_1\varepsilon_{t-s-1} + \beta_2\varepsilon_{t-s-2} + \cdots)^2] \\ &= \sigma^2[1 + (\beta_1)^2 + (\beta_2)^2 + \cdots]\end{aligned}$$

Thus, $\text{var}(x_t) = \text{var}(x_{t-s})$ for all t and $t - s$.

3. Are all autocovariances finite and time independent? First, form $E(x_tx_{t-s})$ as

$$E[x_tx_{t-s}] = E[(\varepsilon_t + \beta_1\varepsilon_{t-1} + \beta_2\varepsilon_{t-2} + \cdots)(\varepsilon_{t-s} + \beta_1\varepsilon_{t-s-1} + \beta_2\varepsilon_{t-s-2} + \cdots)]$$

Carrying out the multiplication and noting that $E(\varepsilon_t\varepsilon_{t-s}) = 0$ for $s \neq 0$, we get

$$E(x_tx_{t-s}) = \sigma^2(\beta_s + \beta_1\beta_{s+1} + \beta_2\beta_{s+2} + \cdots)$$

Restricting the sum $\beta_s + \beta_1\beta_{s+1} + \beta_2\beta_{s+2} + \cdots$ to be finite means that $E(x_tx_{t-s})$ is finite. Given this second restriction, it is clear that the covariance between x_t and x_{t-s} only depends on the number of periods separating the variables (i.e., the value of s) but *not* on the time subscript t.

In summary, the necessary and sufficient conditions for any MA process to be covariance stationary are for the sums $\sum(\beta_i)^2$ and $(\beta_s + \beta_1\beta_{s+1} + \beta_2\beta_{s+2} + \cdots)$ to be finite. For an infinite-order process, these conditions must hold for all $s \geq 0$. Some of the details involved with maximum likelihood estimation of MA processes are discussed in Appendix 2.1.

Stationarity Restrictions for the Autoregressive Coefficients

Now consider the pure autoregressive model

$$y_t = a_0 + \sum_{i=1}^{p} a_iy_{t-i} + \varepsilon_t \tag{2.19}$$

If the characteristic roots of the homogeneous equation of (2.19) all lie inside the unit circle, it is possible to write the particular solution as

$$y_t = a_0 \Big/ \left[1 - \sum_{i=1}^{p} a_i\right] + \sum_{i=0}^{\infty} c_i\varepsilon_{t-i} \tag{2.20}$$

where c_i = undetermined coefficients.

Although it is possible to find the undetermined coefficients $\{c_i\}$, we know that (2.20) is a convergent sequence so long as the characteristic roots of (2.19) are inside the unit circle. To sketch the proof, the method of undetermined coefficients allows us to write the particular solution in the form of (2.20). We also know that the sequence $\{c_i\}$ will eventually solve the difference equation

$$c_i - a_1 c_{i-1} - a_2 c_{i-2} - \cdots - a_p c_{i-p} = 0 \tag{2.21}$$

If the characteristic roots of (2.21) are all inside the unit circle, the $\{c_i\}$ sequence will be convergent. Although (2.20) is an infinite-order moving average process, the convergence of the MA coefficients implies that Σc_i^2 is finite. Thus, we can use (2.20) to check the three conditions for stationarity. From (2.20),

$$Ey_t = Ey_{t-s} = a_0 / \left(1 - \sum a_i\right)$$

You should recall from Chapter 1 that a necessary condition for all characteristic roots to lie inside the unit circle is $1 - \Sigma a_i > 0$. Hence, the mean of the sequence is finite and time invariant.

$$\mathrm{Var}(y_t) = E[(\varepsilon_t + c_1\varepsilon_{t-1} + c_2\varepsilon_{t-2} + c_3\varepsilon_{t-3} + \cdots)^2] = \sigma^2 \sum c_i^2$$

and

$$\mathrm{var}(y_{t-s}) = E[(\varepsilon_{t-s} + c_1\varepsilon_{t-s-1} + c_2\varepsilon_{t-s-2} + c_3\varepsilon_{t-s-3} + \cdots)^2] = \sigma^2 \sum c_i^2$$

Given that $\sum c_i^2$ is finite, the variance is finite and time independent.

$$\begin{aligned}\mathrm{Cov}(y_t, y_{t-s}) &= E[(\varepsilon_t + c_1\varepsilon_{t-1} + c_2\varepsilon_{t-2} + \cdots)(\varepsilon_{t-s} + c_1\varepsilon_{t-s-1} + c_2\varepsilon_{t-s-2} + \cdots)] \\ &= \sigma^2(c_s + c_1 c_{s+1} + c_2 c_{s+2} + \cdots)\end{aligned}$$

Thus, the covariance between y_t and y_{t-s} is constant and time invariant for all t and $t - s$. Nothing of substance is changed by combining the AR(p) and MA(q) models into the general ARMA(p, q) model

$$\begin{aligned} y_t &= a_0 + \sum_{i=1}^{p} a_i y_{t-i} + x_t \\ x_t &= \sum_{i=0}^{q} \beta_i \varepsilon_{t-i} \end{aligned} \tag{2.22}$$

If the roots of the inverse characteristic equation lie outside the unit circle [i.e., if the roots of the homogeneous form of (2.22) lie inside the unit circle] and if the $\{x_t\}$ sequence is stationary, the $\{y_t\}$ sequence will be stationary. Consider

$$y_t = \frac{a_0}{1 - \sum_{i=1}^{p} a_i} + \frac{\varepsilon_t}{1 - \sum_{i=1}^{p} a_i L^i} + \frac{\beta_1 \varepsilon_{t-1}}{1 - \sum_{i=1}^{p} a_i L^i} + \frac{\beta_2 \varepsilon_{t-2}}{1 - \sum_{i=1}^{p} a_i L^i} + \cdots \tag{2.23}$$

With very little effort, you can convince yourself that the $\{y_t\}$ sequence satisfies the three conditions for stationarity. Each of the expressions on the right-hand side of (2.23) is stationary as long as the roots of $1 - \Sigma a_i L^i$ are outside the unit circle. Given that $\{x_t\}$ is stationary, only the roots of the autoregressive portion of (2.22) determine whether the $\{y_t\}$ sequence is stationary.

5. THE AUTOCORRELATION FUNCTION

The autocovariances and autocorrelations of the type found in (2.18) serve as useful tools in the Box–Jenkins (1976) approach to identifying and estimating time-series models. We illustrate by considering four important examples: the AR(1), AR(2), MA(1), and ARMA(1, 1) models. For the AR(1) model, $y_t = a_0 + a_1 y_{t-1} + \varepsilon_t$, (2.14) shows

$$\gamma_0 = \sigma^2/[1 - (a_1)^2]$$
$$\gamma_s = \sigma^2(a_1)^s/[1 - (a_1)^2]$$

Forming the autocorrelations by dividing each γ_s by γ_0, we find that $\rho_0 = 1$, $\rho_1 = a_1$, $\rho_2 = (a_1)^2, \ldots, \rho_s = (a_1)^s$. For an AR(1) process, a necessary condition for stationarity is for $|a_1| < 1$. Thus, the plot of ρ_s against s—called the **autocorrelation function (ACF)** or **correlogram**—should converge to zero geometrically if the series is stationary. If a_1 is positive, convergence will be direct, and if a_1 is negative, the autocorrelations will follow a dampened oscillatory path around zero. The first two graphs on the left-hand side of Figure 2.2 show the theoretical autocorrelation functions for $a_1 = 0.7$ and $a_1 = -0.7$, respectively. Here, ρ_0 is not shown since its value is necessarily unity.

The Autocorrelation Function of an AR(2) Process

Now consider the more complicated AR(2) process $y_t = a_1 y_{t-1} + a_2 y_{t-2} + \varepsilon_t$. We omit an intercept term (a_0) since it has no effect on the ACF. For the second-order process to be stationary, we know that it is necessary to restrict the roots of $(1 - a_1 L - a_2 L^2)$ to be outside the unit circle. In Section 4, we derived the autocovariances of an ARMA(2, 1) process by use of the method of undetermined coefficients. Now, we want to illustrate an alternative technique using the **Yule–Walker** equations. Multiply the second-order difference equation by y_{t-s} for $s = 0, s = 1, s = 2, \ldots$ and take expectations to form

$$\begin{aligned}
Ey_t y_t &= a_1 Ey_{t-1} y_t + a_2 Ey_{t-2} y_t + E\varepsilon_t y_t \\
Ey_t y_{t-1} &= a_1 Ey_{t-1} y_{t-1} + a_2 Ey_{t-2} y_{t-1} + E\varepsilon_t y_{t-1} \\
Ey_t y_{t-2} &= a_1 Ey_{t-1} y_{t-2} + a_2 Ey_{t-2} y_{t-2} + E\varepsilon_t y_{t-2} \\
&\ldots \\
&\ldots \\
Ey_t y_{t-s} &= a_1 Ey_{t-1} y_{t-s} + a_2 Ey_{t-2} y_{t-s} + E\varepsilon_t y_{t-s}
\end{aligned} \tag{2.24}$$

By definition, the autocovariances of a stationary series are such that $Ey_t y_{t-s} = Ey_{t-s} y_t = Ey_{t-k} y_{t-k-s} = \gamma_s$. We also know that $E\varepsilon_t y_t = \sigma^2$ and $E\varepsilon_t y_{t-s} = 0$. Hence, we can use the equations in (2.24) to form

$$\gamma_0 = a_1 \gamma_1 + a_2 \gamma_2 + \sigma^2 \tag{2.25}$$

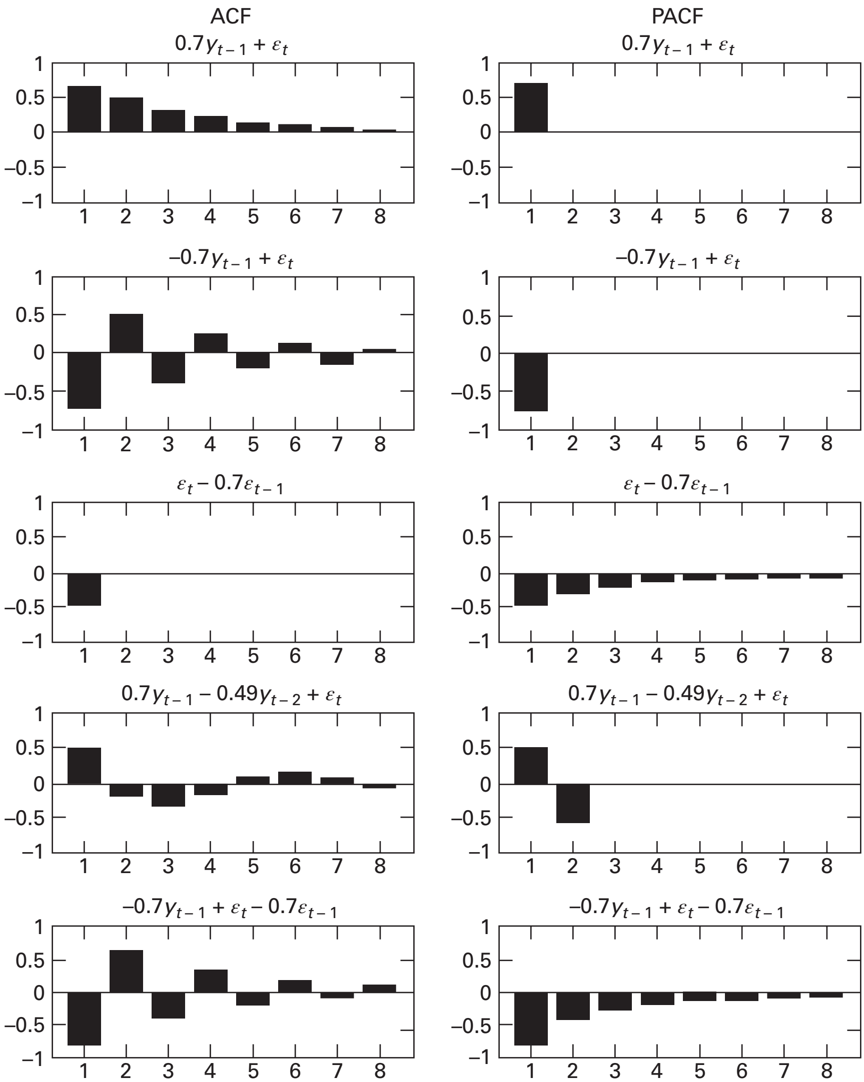


FIGURE 2.2 Theoretical ACF and PACF Patterns

$$\gamma_1 = a_1\gamma_0 + a_2\gamma_1 \tag{2.26}$$

$$\gamma_s = a_1\gamma_{s-1} + a_2\gamma_{s-2} \tag{2.27}$$

Dividing (2.26) and (2.27) by γ_0 yields

$$\rho_1 = a_1\rho_0 + a_2\rho_1 \tag{2.28}$$

$$\rho_s = a_1\rho_{s-1} + a_2\rho_{s-2} \tag{2.29}$$

We know that $\rho_0 = 1$, so that from (2.28), $\rho_1 = a_1/(1 - a_2)$. Hence, we can find all ρ_s for $s \geq 2$ by solving the difference equation (2.29). For example, for $s = 2$

and $s = 3$,

$$\rho_2 = (a_1)^2/(1 - a_2) + a_2$$
$$\rho_3 = a_1[(a_1)^2/(1 - a_2) + a_2] + a_2 a_1/(1 - a_2)$$

Although the values of the ρ_s are cumbersome to derive, we can easily characterize their properties. Given the solutions for ρ_0 and ρ_1, the key point to note is that the ρ_s all satisfy the difference equation (2.29). As in the case of a second-order difference equation, the solution may be oscillatory or direct. Note that the stationarity condition for y_t necessitates that the characteristic roots of (2.29) lie inside the unit circle. Hence, the $\{\rho_s\}$ sequence must be convergent. The correlogram for an AR(2) process must be such that $\rho_0 = 1$ and ρ_1 be determined by (2.28). These two values can be viewed as *initial values* for the second-order difference equation (2.29).

The fourth panel on the left-hand side of Figure 2.2 shows the ACF for the process $y_t = 0.7y_{t-1} - 0.49y_{t-2} + \varepsilon_t$. The properties of the various ρ_s follow directly from the homogeneous equation $y_t - 0.7y_{t-1} + 0.49y_{t-2} = 0$. The roots are obtained from the solution to

$$\alpha = \{0.7 \pm [(-0.7)^2 - 4(0.49)]^{1/2}\}/2$$

Since the discriminant $d = (-0.7)^2 - 4(0.49)$ is negative, the characteristic roots are imaginary so that the solution oscillates. However, since $a_2 = -0.49$, the solution is convergent and the $\{y_t\}$ sequence is stationary.

Finally, we may wish to find the autocovariances rather than the autocorrelations. Since we know all of the autocorrelations, if we can find the variance of y_t (i.e., γ_0), we can find all of the other γ_s. To find γ_0, use (2.25) and note that $\rho_i = \gamma_i/\gamma_0$ so that

$$\gamma_0(1 - a_1\rho_1 - a_2\rho_2) = \sigma^2$$

Substitution for ρ_1 and ρ_2 yields

$$\gamma_0 = \text{var}(y_t) = [(1 - a_2)/(1 + a_2)]\left(\frac{\sigma^2}{(a_1 + a_2 - 1)(a_2 - a_1 - 1)}\right)$$

The Autocorrelation Function of an MA(1) Process

Next consider the MA(1) process $y_t = \varepsilon_t + \beta\varepsilon_{t-1}$. Again, we can obtain the Yule–Walker equations by multiplying y_t by each y_{t-s} and take expectations

$$\gamma_0 = \text{var}(y_t) = Ey_t y_t = E[(\varepsilon_t + \beta\varepsilon_{t-1})(\varepsilon_t + \beta\varepsilon_{t-1})] = (1 + \beta^2)\sigma^2$$
$$\gamma_1 = Ey_t y_{t-1} = E[(\varepsilon_t + \beta\varepsilon_{t-1})(\varepsilon_{t-1} + \beta\varepsilon_{t-2})] = \beta\sigma^2$$

and

$$\gamma_s = Ey_t y_{t-s} = E[(\varepsilon_t + \beta\varepsilon_{t-1})(\varepsilon_{t-s} + \beta\varepsilon_{t-s-1})] = 0 \qquad \text{for all } s > 1$$

Hence, dividing each γ_s by γ_0, it can be immediately seen that the ACF is simply $\rho_0 = 1$, $\rho_1 = \beta/(1 + \beta^2)$, and $\rho_s = 0$ for all $s > 1$. The third graph on the left-hand side of Figure 2.2 shows the ACF for the MA(1) process $y_t = \varepsilon_t - 0.7\,\varepsilon_{t-1}$. As an exercise, you should demonstrate that the ACF for an MA(2) process has two spikes and then cuts to zero.

The Autocorrelation Function of an ARMA(1,1) Process

Finally, let $y_t = a_1 y_{t-1} + \varepsilon_t + \beta_1 \varepsilon_{t-1}$. Using the now-familiar procedure, we find the Yule–Walker equations

$$Ey_t y_t = a_1 Ey_{t-1} y_t + E\varepsilon_t y_t + \beta_1 E\varepsilon_{t-1} y_t \Rightarrow \gamma_0 = a_1 \gamma_1 + \sigma^2 + \beta_1 (a_1 + \beta_1)\sigma^2 \quad (2.30)$$

$$Ey_t y_{t-1} = a_1 Ey_{t-1} y_{t-1} + E\varepsilon_t y_{t-1} + \beta_1 E\varepsilon_{t-1} y_{t-1} \Rightarrow \gamma_1 = a_1 \gamma_0 + \beta_1 \sigma^2 \quad (2.31)$$

$$Ey_t y_{t-2} = a_1 Ey_{t-1} y_{t-2} + E\varepsilon_t y_{t-2} + \beta_1 E\varepsilon_{t-1} y_{t-2} \Rightarrow \gamma_2 = a_1 \gamma_1 \quad (2.32)$$

$$\vdots$$

$$Ey_t y_{t-s} = a_1 Ey_{t-1} y_{t-s} + E\varepsilon_t y_{t-s} + \beta_1 E\varepsilon_{t-1} y_{t-s} \Rightarrow \gamma_s = a_1 \gamma_{s-1} \quad (2.33)$$

In obtaining (2.30), note that $E\varepsilon_{t-1} y_t$ is $(a_1 + \beta_1)\sigma^2$. (2.30) and (2.31) simultaneously for γ_0 and γ_1 yields

$$\gamma_0 = \frac{1 + \beta_1^2 + 2a_1\beta_1}{(1 - a_1^2)}\sigma^2$$

$$\gamma_1 = \frac{(1 + a_1\beta_1)\,(a_1 + \beta_1)}{(1 - a_1^2)}\sigma^2$$

Form the ratio γ_1/γ_0 to obtain

$$\rho_1 = \frac{(1 + a_1\beta_1)\,(a_1 + \beta_1)}{(1 + \beta_1^2 + 2a_1\beta_1)} \quad (2.34)$$

and $\rho_s = a_1 \rho_{s-1}$ for all $s \geq 2$.

Thus, the ACF for an ARMA(1, 1) process is such that the magnitude of ρ_1 depends on both a_1 and β_1. Beginning with this value of ρ_1, the ACF of an ARMA(1, 1) process looks like that of the AR(1) process. If $0 < a_1 < 1$, convergence will be direct, and if $-1 < a_1 < 0$, the autocorrelations will oscillate. The ACF for the function $y_t = -0.7y_{t-1} + \varepsilon_t - 0.7\,\varepsilon_{t-1}$ is shown as the last graph on the left-hand side of Figure 2.2. The top portion of Worksheet 2.2 derives these autocorrelations.

We leave you with the exercise of deriving the correlogram of the ARMA(2, 1) process used in Worksheet 2.1. You should be able to recognize the point that the correlogram can reveal the pattern of the autoregressive coefficients. For an ARMA (p, q) model beginning after lag q, the values of the ρ_i will satisfy

$$\rho_i = a_1\rho_{i-1} + a_2\rho_{i-2} + \cdots + a_p\rho_{i-p}$$

The previous p values can be treated as initial conditions that satisfy the Yule–Walker equations. For these lags, the shape of the ACF is determined by the characteristic equation.

WORKSHEET 2.2

CALCULATION OF THE PARTIAL AUTOCORRELATIONS OF $y_t = -0.7y_{t-1} + \varepsilon_t - 0.7\varepsilon_{t-1}$

Step 1: Calculate the autocorrelations. Use (2.34) to calculate ρ_1 as

$$\rho_1 = \frac{(1+0.49)(-0.7 - 0.7)}{1 + 0.49 + 2(0.49)} = -0.8445$$

The remaining autocorrelations decay at the rate $\rho_i = -0.7\rho_{i-1}$ so that

$$\rho_2 = 0.591,\ \rho_3 = -0.414,\ \rho_4 = 0.290,\ \rho_5 = -0.203,\ \rho_6 = 0.142,\ \rho_7 = -0.099,\ \rho_8 = 0.070$$

Step 2: Calculate the first two partial autocorrelations using (2.35) and (2.36). Hence,

$$\phi_{11} = \rho_1 = -0.8445$$
$$\phi_{22} = [0.591 - (-0.8445)^2]/[1 - (-0.8445)^2] = -0.426$$

Step 3: Calculate all remaining ϕ_{ss} iteratively using (2.37). To find ϕ_{33}, note that $\phi_{21} = \phi_{11} - \phi_{22}\phi_{11} = -1.204$ and form

$$\begin{aligned}\phi_{33} &= \left(\rho_3 - \sum_{j=1}^{2} \phi_{2j}\rho_{3-j}\right)\left(1 - \sum_{j=1}^{2} \phi_{2j}\rho_j\right)^{-1} \\ &= [-0.414 - (-1.204)(0.591) - (-0.426)(-0.8445)]/[1 - (-1.204)(-0.8445) \\ &\qquad -(-0.426)(0.591)] \\ &= -0.262\end{aligned}$$

Similarly, to find ϕ_{44}, use

$$\phi_{44} = \left(\rho_4 - \sum_{j=1}^{3} \phi_{3j}\rho_{4-j}\right)\left(1 - \sum_{j=1}^{3} \phi_{3j}\rho_j\right)^{-1}$$

Since $\phi_{3j} = \phi_{2j} - \phi_{33}\phi_{2,2-j}$, it follows that $\phi_{31} = -1.315$ and $\phi_{32} = -0.74$. Hence,

$$\phi_{44} = -0.173$$

If we continue in this manner, it is possible to demonstrate that $\phi_{55} = -0.117$, $\phi_{66} = -0.081$, $\phi_{77} = -0.056$, and $\phi_{88} = -0.039$.

6. THE PARTIAL AUTOCORRELATION FUNCTION

In an AR(1) process, y_t and y_{t-2} are correlated even though y_{t-2} does not directly appear in the model. The correlation between y_t and y_{t-2} (i.e., ρ_2) is equal to the correlation between y_t and y_{t-1} (i.e., ρ_1) multiplied by the correlation between y_{t-1} and y_{t-2} (i.e., ρ_1 again) so that $\rho_2 = (\rho_1)^2$. It is important to note that all such *indirect* correlations are present in the ACF of any autoregressive process. In contrast, the **partial autocorrelation** between y_t and y_{t-s} eliminates the effects of the intervening values y_{t-1}

through y_{t-s+1}. As such, in an AR(1) process, the partial autocorrelation between y_t and y_{t-2} is equal to zero. The most direct way to find the partial autocorrelation function is to first form the series $\{y_t^*\}$ by subtracting the mean of the series (i.e., μ) from each observation to obtain $y_t^* \equiv y_t - \mu$. Next, form the first-order autoregression

$$y_t^* = \phi_{11} y_{t-1}^* + e_t$$

where e_t is an error term.

Here, the symbol $\{e_t\}$ is used since this error process may not be white noise.

Since there are no intervening values, ϕ_{11} is both the autocorrelation and the partial autocorrelation between y_t and y_{t-1}. Now, form the second-order autoregression equation

$$y_t^* = \phi_{21} y_{t-1}^* + \phi_{22} y_{t-2}^* + e_t$$

Here, ϕ_{22} is the partial autocorrelation coefficient between y_t and y_{t-2}. In other words, ϕ_{22} is the correlation between y_t and y_{t-2} controlling for (i.e., "netting out") the effect of y_{t-1}. Repeating this process for all additional lags s yields the partial autocorrelation function (PACF). In practice, with sample size T, only $T/4$ lags are used in obtaining the sample PACF.

Since most statistical computer packages perform these transformations, there is little need to elaborate on the computational procedure. However, it should be pointed out that a simple computational method relying on the so-called Yule–Walker equations is available. One can form the partial autocorrelations from the autocorrelations as

$$\phi_{11} = \rho_1 \tag{2.35}$$

$$\phi_{22} = (\rho_2 - \rho_1^2)/(1 - \rho_1^2) \tag{2.36}$$

and, for additional lags,

$$\phi_{ss} = \frac{\rho_s - \sum_{j=1}^{s-1} \phi_{s-1,j}\rho_{s-j}}{1 - \sum_{j=1}^{s-1} \phi_{s-1,j}\rho_j}, \qquad s = 3, 4, 5, \ldots. \tag{2.37}$$

where $\phi_{sj} = \phi_{s-1,j} - \phi_{ss}\phi_{s-1,s-j}$, $j = 1, 2, 3, \ldots, s-1$.

For an AR(p) process, there is no direct correlation between y_t and y_{t-s} for $s > p$. Hence, for $s > p$, all values of ϕ_{ss} will be zero, and the PACF for a pure AR(p) process should cut to zero for all lags greater than p. This is a useful feature of the PACF that can aid in the identification of an AR(p) model. In contrast, consider the PACF for the MA(1) process: $y_t = \varepsilon_t + \beta\varepsilon_{t-1}$. As long as $\beta \neq -1$, we can write $y_t/(1 + \beta L) = \varepsilon_t$, which we know has the infinite-order autoregressive representation

$$y_t - \beta y_{t-1} + \beta^2 y_{t-2} - \beta^3 y_{t-3} + \cdots = \varepsilon_t$$

As such, the PACF will *not* jump to zero since y_t will be correlated with all of its own lags. Instead, the PACF coefficients exhibit a geometrically decaying pattern. If $\beta < 0$, decay is direct, and if $\beta > 0$, the PACF coefficients oscillate.

Worksheet 2.2 illustrates the procedure used in constructing the PACF for the ARMA(1, 1) model shown in the fifth panel on the right-hand side of Figure 2.2:

$$y_t = -0.7y_{t-1} + \varepsilon_t - 0.7\,\varepsilon_{t-1}$$

First, calculate the autocorrelations. Clearly, $\rho_0 = 1$; use equation (2.34) to calculate as $\rho_1 = -0.8445$. Thereafter, the ACF coefficients decay at the rate $\rho_i = (-0.7)\rho_{i-1}$ for $i \geq 2$. Using (2.35) and (2.36), we obtain $\phi_{11} = -0.8445$ and $\phi_{22} = -0.4250$. All subsequent ϕ_{ss} and ϕ_{sj} can be calculated from (2.37) as in Worksheet 2.2.

More generally, the PACF of a stationary ARMA(p, q) process must ultimately decay toward zero beginning at lag p. The decay pattern depends on the coefficients of the polynomial $(1 + \beta_1 L + \beta_2 L^2 + \cdots + \beta_q L^q)$. Table 2.1 summarizes some of the properties of the ACF and PACF for various ARMA processes. Also, the right-hand side graphs of Figure 2.2 show the partial autocorrelation functions of the five indicated processes.

For stationary processes, the key points to note are the following:

1. The ACF of an ARMA(p, q) process will begin to decay after lag q. After lag q, the coefficients of the ACF (i.e., the ρ_i) will satisfy the difference equation $(\rho_i = a_1\rho_{i-1} + a_2\rho_{i-2} + \cdots + a_p\rho_{i-p})$. Since the characteristic roots are inside the unit circle, the autocorrelations will decay after lag q. Moreover, the pattern of the autocorrelation coefficients will mimic that suggested by the characteristic roots.
2. The PACF of an ARMA(p, q) process will begin to decay after lag p. After lag p, the coefficients of the PACF (i.e., the ϕ_{ss}) will mimic the ACF coefficients from the model $y_t/(1 + \beta_1 L + \beta_2 L^2 + \cdots + \beta_q L^q)$.

Table 2.1 Properties of the ACF and PACF

Process	**ACF**	**PACF**
White noise	All $\rho_s = 0$ $(s \neq 0)$	All $\phi_{ss} = 0$
AR(1): $a_1 > 0$	Direct geometric decay: $\rho_s = a_1^s$	$\phi_{11} = \rho_1$; $\phi_{ss} = 0$ for $s \geq 2$
AR(1): $a_1 < 0$	Oscillating decay: $\rho_s = a_1^s$	$\phi_{11} = \rho_1$; $\phi_{ss} = 0$ for $s \geq 2$
AR(p)	Decays toward zero. Coefficients may oscillate.	Spikes through lag p. All $\phi_{ss} = 0$ for $s > p$
MA(1): $\beta > 0$	Positive spike at lag 1. $\rho_s = 0$ for $s \geq 2$	Oscillating decay: $\phi_{11} > 0$
MA(1): $\beta < 0$	Negative spike at lag 1. $\rho_s = 0$ for $s \geq 2$	Geometric decay: $\phi_{11} < 0$
ARMA(1, 1): $a_1 > 0$	Geometric decay beginning after lag 1. Sign $\rho_1 = \text{sign}(a_1 + \beta)$	Oscillating decay after lag 1. $\phi_{11} = \rho_1$
ARMA(1, 1): $a_1 < 0$	Oscillating decay beginning after lag 1. Sign $\rho_1 = \text{sign}(a_1 + \beta)$	Geometric decay beginning after lag 1. $\phi_{11} = \rho_1$ and $\text{sign}(\phi_{ss}) = \text{sign}(\phi_{11})$
ARMA(p, q)	Decay (either direct or oscillatory) beginning after lag q	Decay (either direct or oscillatory) beginning after lag p

We can illustrate the usefulness of the ACF and PACF functions using the model $y_t = a_0 + 0.7y_{t-1} + \varepsilon_t$. If we compare the top two graphs in Figure 2.2, the ACF shows the monotonic decay of the autocorrelations while the PACF exhibits the single spike at lag 1. Suppose that a researcher collected sample data and plotted the ACF and PACF. If the actual patterns compared favorably to the theoretical patterns, the researcher might try to estimate data using an AR(1) model. Correspondingly, if the ACF exhibited a single spike and the PACF exhibited monotonic decay (see the third graph for the model $y_t = \varepsilon_t - 0.7\,\varepsilon_{t-1}$), the researcher might try an MA(1) model.

7. SAMPLE AUTOCORRELATIONS OF STATIONARY SERIES

In practice, the theoretical mean, variance, and autocorrelations of a series are unknown to the researcher. Given that a series is stationary, we can use the sample mean, variance, and autocorrelations to estimate the parameters of the actual data-generating process. Let there be T observations labeled y_1 through y_T. We can let $\bar{y}$, $\hat{\sigma}^2$, and r_s be estimates of μ, σ^2, and ρ_s, respectively where[1]

$$\bar{y} = (1/T)\sum_{t=1}^{T} y_t \tag{2.38}$$

$$\hat{\sigma}^2 = (1/T)\sum_{t=1}^{T} (y_t - \bar{y})^2 \tag{2.39}$$

and for each value of $s = 1, 2, \ldots,$

$$r_s = \frac{\sum_{t=s+1}^{T} (y_t - \bar{y})(y_{t-s} - \bar{y})}{\sum_{t=1}^{T} (y_t - \bar{y})^2} \tag{2.40}$$

The sample autocorrelation function [i.e., the ACF derived from (2.40)] and the sample PACF can be compared to various theoretical functions to help identify the actual nature of the data-generating process. Box and Jenkins (1976) discuss the distribution of the sample values of r_s under the null that y_t is stationary with normally distributed errors. Allowing $\mathrm{var}(r_s)$ to denote the sampling variance of r_s, they obtained

$$\begin{aligned} \mathrm{var}(r_s) &= T^{-1} && \text{for } s = 1 \\ &= T^{-1}\left(1 + 2\sum_{j=1}^{s-1} r_j^2\right) && \text{for } s > 1 \end{aligned} \tag{2.41}$$

if the true value of $r_s = 0$ [i.e., if the true data-generating process is an MA$(s-1)$ process]. Moreover, in large samples (i.e., for large values of T), r_s will be normally distributed with a mean equal to zero. For the PACF coefficients, under the null hypothesis of an AR(p) model (i.e., under the null that all $\phi_{p+i,p+i}$ are zero), the variance of the $\hat{\phi}_{p+i,p+i}$ is approximately $1/T$ (See Section 2.3 of the *Supplementary Manual*).

In practice, we can use these sample values to form the sample autocorrelations and partial autocorrelation functions and test for significance using (2.41). For example, if we use a 95% confidence interval (i.e., two standard deviations), and the calculated value of r_1 exceeds $2T^{-1/2}$, it is possible to reject the null hypothesis that the first-order autocorrelation is not statistically different from zero. Rejecting this hypothesis means rejecting an MA($s-1$) = MA(0) process and accepting the alternative $q > 0$. Next, try $s = 2$; var(r_2) is $(1 + 2r_1^2)/T$. If r_1 is 0.5 and T is 100, the variance of r_2 is 0.015 and the standard deviation is about 0.123. Thus, if the calculated value of r_2 exceeds 2(0.123), it is possible to reject the hypothesis $r_2 = 0$. Here, rejecting the null means accepting the alternative that $q > 1$. Repeating for the various values of s is helpful in identifying the order to the process. The maximum number of sample autocorrelations and partial autocorrelations to use is typically set equal to $T/4$.

Within any large group of autocorrelations, some will exceed two standard deviations as a result of pure chance even though the true values in the data-generating process are zero. The Q-statistic can be used to test whether a group of autocorrelations is significantly different from zero. Box and Pierce (1970) used the sample autocorrelations to form the statistic

$$Q = T\sum_{k=1}^{s} r_k^2$$

Under the null hypothesis that all values of $r_k = 0$, Q is asymptotically χ^2 distributed with s degrees of freedom. The intuition behind the use of the statistic is that high sample autocorrelations lead to large values of Q. Certainly, a white-noise process (in which all autocorrelations should be zero) would have a Q value of zero. If the calculated value of Q exceeds the appropriate value in a χ^2 table, we can reject the null of no significant autocorrelations. Note that rejecting the null means accepting an alternative that at least one autocorrelation is not zero.

A problem with the Box–Pierce Q-statistic is that it works poorly even in moderately large samples. Ljung and Box (1978) reported superior small sample performance for the modified Q-statistic calculated as

$$Q = T(T+2)\sum_{k=1}^{s} r_k^2/(T-k) \tag{2.42}$$

If the sample value of Q calculated from (2.42) exceeds the critical value of χ^2 with s degrees of freedom, then *at least* one value of r_k is statistically different from zero at the specified significance level. The Box–Pierce and Ljung–Box Q-statistics also serve as a check to see if the *residuals* from an estimated ARMA(p, q) model behave as a white-noise process. However, when the s correlations from an estimated ARMA(p, q) model are formed, the degrees of freedom are reduced by the number of estimated coefficients. Hence, using the residuals of an ARMA(p, q) model, Q has a χ^2 with $s - p - q$ degrees of freedom (if a constant is included, the degrees of freedom are $s - p - q - 1$).

Model Selection Criteria

One natural question to ask of any estimated model is: How well does it fit the data? Adding additional lags for p and/or q will necessarily reduce the sum of squares of the estimated residuals. However, adding such lags entails the estimation of additional coefficients and an associated loss of degrees of freedom. Moreover, the inclusion of extraneous coefficients will reduce the forecasting performance of the fitted model. As discussed in Appendix 2.2 in the *Supplementary Manual*, there exist various model selection criteria that trade-off a reduction in the sum of squares of the residuals for a more **parsimonious** model. The two most commonly used model selection criteria are the Akaike Information Criterion (AIC) and the Schwartz Bayesian Criterion (SBC). In the text, we use the following formulas

$$\text{AIC} = T \ \ln(\text{sum of squared residuals}) + 2n$$
$$\text{SBC} = T \ \ln(\text{sum of squared residuals}) + n \ \ln(T)$$

where n = number of parameters estimated ($p + q$ + possible constant term)

T = number of usable observations.

When you estimate a model using lagged variables, some observations are lost. To adequately compare the alternative models, T should be kept fixed. Otherwise, you will be comparing the performance of the models over different sample periods. Moreover, decreasing T has direct effect of reducing the AIC and the SBC; the goal is not to select a model because it has the smallest number of usable observations. For example, with 100 data points, estimate an AR(1) and an AR(2) using only the last 98 observations in each estimation. Compare the two models using $T = 98$.

Ideally, the AIC and SBC will be as small as possible (note that both can be negative). As the fit of the model improves, the AIC and SBC will approach $-\infty$. We can use these criteria to aid in selecting the most appropriate model; model A is said to fit better than model B if the AIC (or SBC) for A is smaller than for model B. In using the criteria to compare alternative models, we must estimate them over the same sample period so that they will be comparable. For each, increasing the number of regressors increases n but should have the effect of reducing the sum of squared residuals (SSR). Thus, if a regressor has no explanatory power, adding it to the model will cause both the AIC and SBC to increase. Since $\ln(T)$ will be greater than 2, the SBC will always select a more parsimonious model than will the AIC; the marginal cost of adding regressors is greater with the SBC than with the AIC.

An especially useful feature of the model selection criteria is for comparing non-nested models. For example, suppose you want to compare an AR(2) model to an MA(3) model. Neither is a restricted form of the other. You would not want to estimate an ARMA(2, 3) model and perform F-tests to determine whether $a_1 = a_2 = 0$ or whether $\beta_1 = \beta_2 = \beta_3 = 0$. As discussed in Appendix 2.1, the estimation of ARMA models necessitates computer-based solution methods. If the AR(2) and MA(3) models are each reasonable, the nonlinear search algorithms required to estimate an ARMA(2, 3) model are not likely to converge to a solution. Moreover, the values of y_{t-1}

and y_{t-2} are clearly correlated with the values of ε_{t-1}, ε_{t-2}, and ε_{t-3}. It is quite possible that both the hypotheses could be accepted (or rejected). However, it is straightforward to compare the estimated AR(2) and MA(3) models using the AIC or the SBC.

Of the two criteria, the SBC has superior large sample properties. Let the true order of the data-generating process be (p^*, q^*) and suppose that we use the AIC and SBC to estimate all ARMA models of order (p, q) where $p \geq p^*$ and $q \geq q^*$. Both the AIC and the SBC will select models of orders greater than or equal to (p^*, q^*) as the sample size approaches infinity. However, the SBC is asymptotically consistent while the AIC is biased toward selecting an overparameterized model. However, in small samples, the AIC can work better than the SBC. You can be quite confident in your results if both the AIC and the SBC select the same model. If they select different models, you need to proceed cautiously. Since SBC selects the more parsimonious model, you should check to determine if the residuals appear to be white noise. Since the AIC can select an overparameterized model, the t-statistics of all coefficients should be significant at conventional levels. A number of other diagnostic checks that can be used to compare alternative models are presented in Sections 8 and 9. Nevertheless, it is wise to retain a healthy skepticism of your estimated models. With many data sets, it is just not possible to find the one model that clearly dominates all others. There is nothing wrong with reporting the results and the forecasts using alternative estimations.

Before proceeding, be aware that a number of different ways are used to report the AIC and the SBC. For example, the software packages EViews and SAS report values for the AIC and SBC using

$$\text{AIC}^* = -2\ln(L)/T + 2n/T$$
$$\text{SBC}^* = -2\ln(L)/T + n\ \ln(T)/T$$

where n and T are as defined above and L is the maximized value of the log of the likelihood function.

For a normal distribution, $-2\ln(L) = T\ln(2\pi) + T\ln(\sigma^2) + (1/\sigma^2)\ (\text{SSR})$. The reason for the plethora of reporting methods is that many software packages (such as OX, RATS, and GAUSS) do not display any model selection criteria so that users must calculate these values by themselves. Programmers quickly find that coding all of the parameters contained in the formulas is unnecessary and simply report the shortened versions. In point of fact, it does not matter which method you use. If you work through Question 7 at the end of this chapter, it should be clear that the model with the smallest value for AIC will always have the smallest AIC*. Specifically, Question 7 asks you to write down the formula for $\ln(L)$ and show that the equation for AIC* is a monotonic transformation of that for AIC. Hence, whether you use the formula for AIC or AIC*, you will always be selecting the same model as the one selected in the text. The identical relationship holds between SBC* and SBC; the model yielding the smallest value for SBC will always have the smallest value for SBC*.

Estimation of an AR(1) Model

Let us use a specific example to see how the sample autocorrelation function and partial autocorrelation function can be used as an aid in identifying an ARMA model. A computer program was used to draw 100 normally distributed random numbers with a theoretical variance equal to unity. Call these random variates ε_t, where t runs

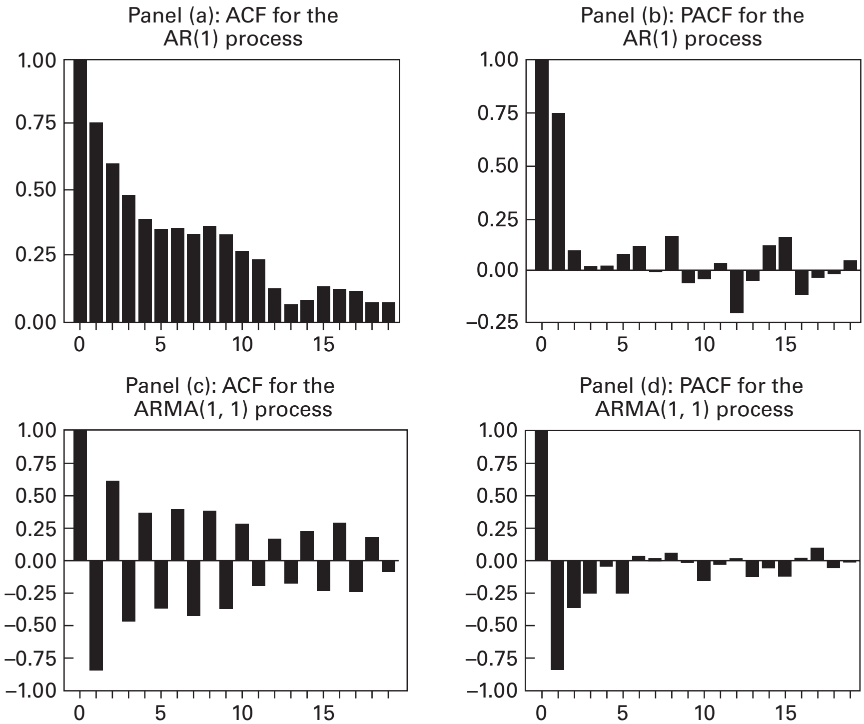


FIGURE 2.3 ACF and PACF for Two Simulated Processes

from 1 to 100. Beginning with $t = 1$, values of y_t were generated using the formula $y_t = 0.7y_{t-1} + \varepsilon_t$ and the initial condition $y_0 = 0$. Note that the problem of nonstationarity is avoided since the initial condition is consistent with long-run equilibrium. Panel (a) of Figure 2.3 shows the sample correlogram and Panel (b) shows the sample PACF. You should take a minute to compare the ACF and PACF to those of the theoretical processes shown in Figure 2.2.

In practice, we never know the true data-generating process. As an exercise, suppose we were presented with these 100 sample values and were asked to uncover the true process using the Box–Jenkins methodology. The first step might be to compare the sample ACF and PACF to those of the various theoretical models. The decaying pattern of the ACF and the single large spike at lag 1 in the sample PACF suggests an AR(1) model. The first three autocorrelations are $r_1 = 0.74$, $r_2 = 0.58$, and $r_3 = 0.47$. [Note that these are somewhat greater than the theoretical values of 0.7, 0.49 ($0.7^2 = 0.49$), and 0.343, respectively.] In the PACF, there is a sizable spike of 0.74 at lag 1, and all other partial autocorrelations (except for lag 12) are very small.

Under the null hypothesis of an MA(0) process, the standard deviation of r_1 is $T^{-1/2} = 0.1$. Since the sample value of $r_1 = 0.74$ is more than seven standard deviations from zero, we can reject the null hypothesis that r_1 equals 0. The standard deviation of r_2 is obtained by applying (2.41) to the sampling data, where $s = 2$

$$\text{var}(r_2) = (1 + 2(0.74)^2)/100 = 0.021$$

Since $(0.021)^{1/2} = 0.1449$, the sample value of r_2 is more than three standard deviations from zero; at conventional significance levels, we can reject the null hypothesis that r_2 equals zero. We can similarly test the significance of the other values of the autocorrelations.

As you can see in Panel (b) of the figure, other than ϕ_{11}, all partial autocorrelations (except for lag 12) are less than $2T^{-1/2} = 0.2$. The decay of the ACF and the single spike of the PACF give the strong impression of a first-order autoregressive model. Nevertheless, if we did not know the true underlying process and happened to be using monthly data, we might be concerned with the significant partial autocorrelation at lag 12. After all, with monthly data, we might expect some direct relationship between y_t and y_{t-12}.

Although we know that the data was actually generated from an AR(1) process, it is illuminating to compare the estimates of two different models. Suppose we estimate an AR(1) model and try to capture the spike at lag 12 with an MA coefficient. Thus, we can consider the two tentative models

$$\text{Model 1: } y_t = a_1 y_{t-1} + \varepsilon_t$$
$$\text{Model 2: } y_t = a_1 y_{t-1} + \varepsilon_t + \beta_{12}\varepsilon_{t-12}.$$

Table 2.2 reports the results of the two estimations. The coefficient of model 1 satisfies the stability condition $|a_1| < 1$ and has a low standard error (the associated t-statistic for a null of zero is more than 12). As a useful diagnostic check, we plot the correlogram of the **residuals** of the fitted model in Figure 2.4. The Q-statistics for these residuals indicate that each one of the autocorrelations is less than two standard deviations from zero. The Ljung–Box Q-statistics of these residuals indicate that, *as a group*, lags 1 through 8, 1 through 16, and 1 through 24 are not significantly different from zero. This is strong evidence that the AR(1) model "fits" the data well. After all, if residual autocorrelations were significant, the AR(1) model would not use all available information concerning movements in the $\{y_t\}$ sequence. For example, suppose we wanted to forecast y_{t+1} conditioned on all available information up to and including period t. With model 1, the value of y_{t+1} is $y_{t+1} = a_1 y_t + \varepsilon_{t+1}$. Hence, the forecast from

Table 2.2 Estimates of an AR(1) Model

	Model 1 $y_t = a_1 y_{t-1} + \varepsilon_t$	**Model 2** $y_t = a_1 y_{t-1} + \varepsilon_t + \beta_{12}\varepsilon_{t-12}$
Degrees of freedom	98	97
Sum of squared residuals	85.10	85.07
Estimated a_1	0.7904	0.7938
(standard error)	(0.0624)	(0.0643)
Estimated β		−0.0325
(standard error)		(0.1141)
AIC/SBC	AIC = 441.9; SBC = 444.5	AIC = 443.9; SBC = 449.1
Ljung–Box Q-statistics for the residuals (significance level in parentheses)	$Q(8) = 6.43$ (0.490) $Q(16) = 15.86$ (0.391) $Q(24) = 21.74$ (0.536)	$Q(8) = 6.48$ (0.485) $Q(16) = 15.75$ (0.400) $Q(24) = 21.56$ (0.547)

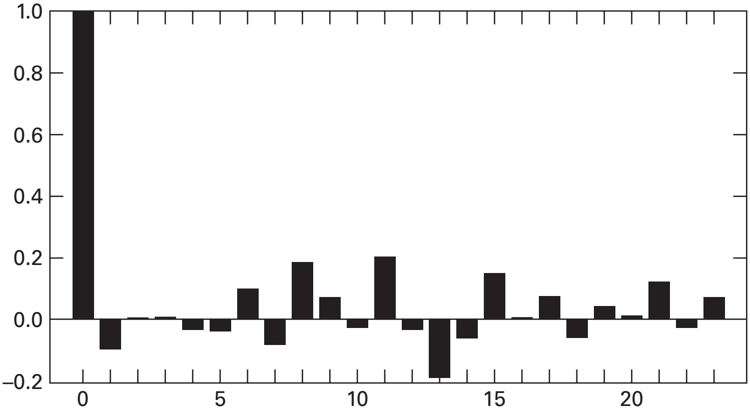


FIGURE 2.4 ACF of Residuals from Model 1

model 1 is $a_1 y_t$. If the residual autocorrelations had been significant, this forecast would not capture all of the available information set.

Examining the results for model 2, note that both models yield similar estimates for the first-order autoregressive coefficient and the associated standard error. However, the estimate for β_{12} is of poor quality; the insignificant t-value suggests that it should be dropped from the model. Moreover, comparing the AIC and the SBC values of the two models suggests that the benefits of reducing the SSR is overwhelmed by the detrimental effects of estimating an additional parameter. All of these indicators point to the choice of model 1.

Exercise 8 at the end of this chapter entails various estimations using this series. The series is denoted by Y1 in the file SIM2.XLS. In this exercise, you are asked to show that the AR(1) model performs better than some alternative specifications. It is important that you complete this exercise.

Estimation of an ARMA(1, 1) Model

A second $\{y_t\}$ sequence in the file SIM2.XLS was constructed to illustrate the estimation of an ARMA(1, 1). Given 100 normally distributed values of $\{\varepsilon_t\}$, 100 values of $\{y_t\}$ were generated using

$$y_t = -0.7y_{t-1} + \varepsilon_t - 0.7\,\varepsilon_{t-1}$$

where y_0 and ε_0 were both set equal to zero.

Both the sample ACF and the PACF from the simulated data (see the second set of graphs in Figure 2.3) are roughly equivalent to those of the theoretical model shown in Figure 2.2. However, if the true data-generating process were unknown, the researcher might be concerned about certain discrepancies. An AR(2) model could yield a sample ACF and PACF similar to those in the figure. Table 2.3 reports the results of estimating the data using the following three models:

$$\text{Model 1: } y_t = a_1 y_{t-1} + \varepsilon_t$$

Table 2.3 Estimates of an ARMA(1,1) Model

	Estimates[1]	*Q*-Statistics[2]	AIC/SBC[3]
Model 1	a_1: −0.835 (0.053)	$Q(8)$ = 26.19 (0.000); $Q(24)$ = 41.10 (0.001)	AIC = 496.5; SBC = 499.0
Model 2	a_1: −0.679 (0.076) β_1: −0.676 (0.081)	$Q(8)$ = 3.86 (0.695); $Q(24)$ = 14.23 (0.892)	AIC = 471.0; SBC = 476.2
Model 3	a_1: −1.16 (0.093) a_2: −0.378 (0.092)	$Q(8)$ = 11.44 (0.057); $Q(24)$ = 22.59 (0.424)	AIC = 482.8; SBC = 487.9

Notes:
[1]Standard errors in parentheses.
[2]Ljung–Box *Q*-statistics of the residuals from the fitted model. The significance levels are in parentheses.
[3]For comparability, the AIC and SBC values are reported for estimations that used only observations 3 through 100. If the AR(1) is estimated using 99 observations, the AIC and SBC are 502.3 and 504.9, respectively. If the ARMA(1, 1) is estimated using 99 observations, the AIC and SBC are 476.6 and 481.1, respectively.

$$\text{Model 2: } y_t = a_1 y_{t-1} + \varepsilon_t + \beta_1 \varepsilon_{t-1}$$
$$\text{Model 3: } y_t = a_1 y_{t-1} + a_2 y_{t-2} + \varepsilon_t$$

In examining Table 2.3, notice that all of the estimated values of a_1 are highly significant; each of the estimated values is *at least* eight standard deviations from zero. It is clear that the AR(1) model is inappropriate. The *Q*-statistics for model 1 indicate that there is significant autocorrelation in the residuals. The estimated ARMA(1, 1) model does not suffer from this problem. Moreover, both the AIC and the SBC select model 2 over model 1.

The same type of reasoning indicates that model 2 is preferred to model 3. Note that, for each model, the estimated coefficients are highly significant and the point estimates imply convergence. Although the *Q*-statistic at 24 lags indicates that these two models do not suffer from correlated residuals, the *Q*-statistic at 8 lags indicates serial correlation in the residuals of model 3. Thus, the AR(2) model does not capture short-term dynamics, as well as the ARMA(1, 1) model. Also note that the AIC and SBC both select model 2.

Estimation of an AR(2) Model

A third data series was simulated as

$$y_t = 0.7y_{t-1} - 0.49y_{t-2} + \varepsilon_t$$

The estimated ACF and PACF of the series are

Lags					Autocorrelations					
1–10	0.47	−0.16	−0.32	−0.11	−0.05	−0.16	−0.10	0.13	0.18	0.03
11–20	−0.09	−0.11	−0.16	−0.06	0.12	0.25	0.05	−0.17	−0.15	0.01
					Partial Autocorrelations					
1–10	0.47	−0.48	0.02	0.05	−0.25	−0.12	0.10	0.04	−0.08	0.02
11–20	−0.02	−0.14	−0.17	0.21	0.01	0.09	−0.22	0.01	−0.02	−0.03

Note the large autocorrelation at lag 16 and the large partial autocorrelations at lags 14 and 17. Given the way the process was simulated, the presence of these autocorrelations is due to nothing more than chance. However, an econometrician unaware of the actual data-generating process might be concerned about these autocorrelations. The estimated AR(2) model (with t-statistics in parentheses) is

$$\underset{}{y_t} = \underset{(7.73)}{0.692 y_{t-1}} - \underset{(-5.37)}{0.481 y_{t-2}} \qquad \text{AIC} = 219.87, \text{SBC} = 225.04$$

Overall, the model appears to be adequate. However, the two AR(2) coefficients are unable to capture the correlations at very long lags. For example, the partial autocorrelations of the *residuals* for lags 14 and 17 are both greater than 0.2 in absolute value. The calculated Ljung–Box statistic for 16 lags is 24.6248 (which is significant at the 0.038 level). At this point, it might be tempting to try to model the correlation at lag 16 by including the moving average term $\beta_{16}\varepsilon_{t-16}$. Such an estimation results in[2]

$$y_t = \underset{(7.87)}{0.717 y_{t-1}} - \underset{(-5.11)}{0.465 y_{t-2}} + \underset{(2.78)}{0.306 \varepsilon_{t-16}} \qquad \text{AIC} = 213.40, \text{SBC} = 221.16$$

All estimated coefficients are significant and the Ljung–Box Q-statistics for the residuals are all insignificant at conventional levels. In conjunction with the fact that the AIC and SBC both select this second model, the researcher unaware of the true process might be tempted to conclude that the data-generating process includes a moving average term at lag 16.

A useful model check is to split the sample into two parts. If a coefficient is present in the data-generating process, its influence should be seen in both subsamples. If the simulated series is split into two parts, the ACF and PACF using observations 50 through 100 follow:

Lags					**Autocorrelations**					
1–10	0.46	−0.21	−0.28	0.03	0.10	−0.15	−0.13	0.10	0.18	0.03
11–20	−0.01	0.01	−0.06	−0.09	0.04	0.21	0.06	−0.16	−0.18	−0.05
					Partial Autocorrelations					
1–10	0.46	−0.53	0.19	0.06	−0.20	−0.13	0.23	−0.08	0.00	0.06
11–20	0.15	−0.26	0.03	0.15	0.04	0.00	−0.05	−0.01	−0.14	−0.08

As you can see, the size of the partial autocorrelations at lags 14 and 17 is diminished. Now, estimating a pure AR(2) model over this second part of the sample yields

$$y_t = \underset{(5.92)}{0.714 y_{t-1}} - \underset{(-4.47)}{0.538 y_{t-2}}$$

$$Q(8) = 7.83; Q(16) = 15.93; Q(24) = 26.06$$

All estimated coefficients are significant, and the Ljung–Box Q-statistics do not indicate any significant autocorrelations in the residuals. The significance levels of $Q(8)$, $Q(16)$, and $Q(24)$ are 0.251, 0.317, and 0.249, respectively. In fact, this model does capture the actual data-generating process quite well. In this example, the large spurious autocorrelations of the long lags can be eliminated by changing the sample period. Thus, it is hard to maintain that the correlation at lag 16 is meaningful. Most sophisticated practitioners warn against trying to fit any model to the very long lags. As you can infer from (2.41), the variance of r_s can be sizable when s is large. Moreover, in small samples, a few "unusual" observations can create the appearance of significant autocorrelations at long lags. Since econometric estimation involves unknown data-generating processes, the more general point is that we always need to be wary of our estimated model. Fortunately, Box and Jenkins (1976) established a set of procedures that can be used to check a model's adequacy.

8. BOX–JENKINS MODEL SELECTION

The estimates of the AR(1), ARMA(1, 1), and AR(2) models in the previous section illustrate the Box–Jenkins (1976) strategy for appropriate model selection. Box and Jenkins popularized a three-stage method aimed at selecting an appropriate model for the purpose of estimating and forecasting a univariate time series. In the **identification stage**, the researcher visually examines the time plot of the series, the autocorrelation function, and the partial correlation function. Plotting the time path of the $\{y_t\}$ sequence provides useful information concerning outliers, missing values, and structural breaks in the data. Nonstationary variables may have a pronounced trend or appear to meander without a constant long-run mean or variance. Missing values and outliers can be corrected at this point. At one time, the standard practice was to first difference any series deemed to be nonstationary. Currently, there is a large body of literature regarding formal procedures to check for nonstationarity. We defer this discussion until Chapter 4 and assume that we are working with stationary data. A comparison of the sample ACF and PACF to those of various theoretical ARMA processes may suggest several plausible models. In the **estimation stage**, each of the tentative models is fit, and the various a_i and β_i coefficients are examined. In this second stage, the goal is to select a stationary and parsimonious model that has a good fit. The third stage involves **diagnostic checking** to ensure that the residuals from the estimated model mimic a white-noise process.

Parsimony

A fundamental idea in the Box–Jenkins approach is the principle of **parsimony**. Parsimony (meaning sparseness or stinginess) should come as second nature to economists. Incorporating additional coefficients will necessarily increase fit (e.g., the value of R^2 will increase) at a cost of reducing degrees of freedom. Box and Jenkins argue that parsimonious models produce better forecasts than overparameterized models. A parsimonious model fits the data well without incorporating any needless coefficients. Certainly, forecasters do not want to project poorly estimated coefficients into the future. The aim is to approximate the true data-generating process but not to pin down the exact process. The goal of parsimony suggested eliminating the MA(12) coefficient in the simulated AR(1) model above.

In selecting an appropriate model, the econometrician needs to be aware that several different models may have similar properties. As an extreme example, note that the AR(1) model $y_t = 0.5y_{t-1} + \varepsilon_t$ has the equivalent infinite-order moving average representation of $y_t = \varepsilon_t + 0.5\varepsilon_{t-1} + 0.25\varepsilon_{t-2} + 0.125\varepsilon_{t-3} + 0.0625\varepsilon_{t-4} + \cdots$. In most samples, approximating this MA(∞) process with an MA(2) or MA(3) model will give a very good fit. However, the AR(1) model is the more parsimonious model and is preferred. As a quiz, you should show that this AR(1) model has the equivalent representation of $y_t = 0.25y_{t-2} + 0.5\varepsilon_{t-1} + \varepsilon_t$.

In addition, be aware of the **common factor problem**. Suppose we wanted to fit the ARMA(2, 3) model

$$(1 - a_1L - a_2L^2)y_t = (1 + \beta_1L + \beta_2L^2 + \beta_3L^3)\varepsilon_t \tag{2.43}$$

Suppose that $(1 - a_1L - a_2L^2)$ and $(1 + \beta_1L + \beta_2L^2 + \beta_3L^3)$ can be factored as $(1 + cL)(1 + aL)$ and $(1 + cL)(1 + b_1L + b_2L^2)$, respectively. Since $(1 + cL)$ is a common factor to each, (2.43) has the equivalent but more parsimonious form:

$$(1 + aL)y_t = (1 + b_1L + b_2L^2)\varepsilon_t \tag{2.44}$$

If you passed the last quiz, you know that $(1 - 0.25L^2)y_t = (1 + 0.5L)\varepsilon_t$ is equivalent to $(1 + 0.5L)(1 - 0.5L)y_t = (1 + 0.5L)\varepsilon_t$ so that $y_t = 0.5y_{t-1} + \varepsilon_t$. In practice, the polynomials will not factor exactly. However, if the factors are similar, you should try a more parsimonious form.

In order to ensure that the model is parsimonious, the various a_i and β_i should all have t-statistics of 2.0 or greater (so that each coefficient is significantly different from zero at the 5% level). Moreover, the coefficients should not be strongly correlated with each other. Highly collinear coefficients are unstable; usually, one or more can be eliminated from the model without reducing forecast performance.

Stationarity and Invertibility

The distribution theory underlying the use of the sample ACF and PACF as approximations to those of the true data-generating process assumes that the $\{y_t\}$ sequence is stationary. Moreover, t-statistics and Q-statistics also presume that the data are stationary. The estimated autoregressive coefficients should be consistent with this underlying assumption. Hence, we should be suspicious of an AR(1) model if the estimated value of a_1 is close to unity. For an ARMA(2, q) model, the characteristic roots of the estimated polynomial $(1 - a_1L - a_2L^2)$ should lie outside of the unit circle.

As discussed in greater detail in Appendix 2.1, the Box–Jenkins approach also necessitates that the model be **invertible**. Formally, $\{y_t\}$ is invertible if it can be represented by a finite-order or convergent autoregressive process. Invertibility is important because the use of the ACF and PACF implicitly assume that the $\{y_t\}$ sequence can be represented by an autoregressive model. As a demonstration, consider the simple MA(1) model:

$$y_t = \varepsilon_t - \beta_1\varepsilon_{t-1} \tag{2.45}$$

so that if $|\beta_1| < 1$,

$$y_t/(1 - \beta_1L) = \varepsilon_t$$

or

$$y_t + \beta_1 y_{t-1} + \beta_1^2 y_{t-2} + \beta_1^3 y_{t-3} + \cdots = \varepsilon_t \tag{2.46}$$

If $|\beta_1| < 1$, (2.46) can be estimated using the Box–Jenkins method. However, if $|\beta_1| \geq 1$, the $\{y_t\}$ sequence cannot be represented by a finite-order AR process; as such, it is not invertible. More generally, for an ARMA model to have a convergent AR representation, the roots of the polynomial $(1 + \beta_1 L + \beta_2 L^2 + \cdots + \beta_q L^q)$ must lie outside the unit circle. Note that there is nothing improper about a noninvertible model. The $\{y_t\}$ sequence implied by $y_t = \varepsilon_t - \varepsilon_{t-1}$ is stationary in that it has a constant time-invariant mean ($Ey_t = Ey_{t-s} = 0$), a constant time-invariant variance [$\text{var}(y_t) = \text{var}(y_{t-s}) = \sigma^2(1 + \beta_1^2) + 2\sigma^2$], and the autocovariances $\gamma_1 = -\beta_1\sigma^2$ and all other $\gamma_s = 0$. The problem is that the technique does not allow for the estimation of such models. If $\beta_1 = 1$, (2.46) becomes

$$y_t + y_{t-1} + y_{t-2} + y_{t-3} + y_{t-4} + \cdots = \varepsilon_t$$

Clearly, the autocorrelations and partial autocorrelations between y_t and y_{t-s} will never decay.

Goodness of Fit

A good model will fit the data well. Obviously, R^2 and the average of the residual sum of squares are common *goodness-of-fit* measures in ordinary least squares. The problem with these measures is that the fit necessarily improves as more parameters are included in the model. Parsimony suggests using the AIC and/or SBC as more appropriate measures of the overall fit of the model. Also, be cautious of estimates that fail to converge rapidly. Most software packages estimate the parameters of an ARMA model using a nonlinear search procedure. If the search fails to converge rapidly, it is possible that the estimated parameters are unstable. In such circumstances, adding an additional observation or two can greatly alter the estimates.

Postestimation Evaluation

The third stage of the Box–Jenkins methodology involves **diagnostic checking**. The standard practice is to plot the residuals to look for outliers and evidence of periods in which the model does not fit the data well. One common practice is to create the standardized residuals by dividing each residual, ε_t, by its estimated standard deviation, σ. If the residuals are normally distributed, the plot of the ε_t/σ series should be such that no more than 5% lie outside the band from −2 to +2. If the standardized residuals seem to be much larger in some periods than in others, it may be evidence of structural change. If all plausible ARMA models show evidence of a poor fit during a reasonably long portion of the sample, it is wise to consider using intervention analysis, transfer function analysis, or any other of the multivariate estimation methods discussed in later chapters. If the variance of the residuals is increasing, a logarithmic transformation may be appropriate. Alternatively, you may wish to actually model any tendency of the variance to change using the ARCH techniques discussed in Chapter 3.

It is particularly important that the residuals from an estimated model be serially uncorrelated. Any evidence of serial correlation implies a systematic movement in the $\{y_t\}$ sequence that is not accounted for by the ARMA coefficients included in the model. Hence, any of the tentative models yielding nonrandom residuals should be eliminated from consideration. To check for correlation in the residuals, construct the ACF and the PACF of the *residuals* of the estimated model. You can then use (2.41) and (2.42) to determine whether any or all of the residual autocorrelations or partial autocorrelations are statistically significant.[3] Although there is no significance level that is deemed "most appropriate," be wary of any model yielding (1) several residual correlations that are marginally significant and (2) a Q-statistic that is barely significant at the 10% level. In such circumstances, it is usually possible to formulate a better performing model.

Similarly, a model can be estimated over only a portion of the data set. The estimated model can then be used to forecast the known values of the series. The sum of the squared forecast errors is a useful way to compare the adequacy of alternative models. Those models with poor *out-of-sample* forecasts should be eliminated. Some of the details in constructing out-of-sample forecasts are discussed in Section 9.

9. PROPERTIES OF FORECASTS

Perhaps the most important use of an ARMA model is to forecast future values of the $\{y_t\}$ sequence. To simplify the discussion, it is assumed that the actual data-generating process and the current and past realizations of the $\{\varepsilon_t\}$ and $\{y_t\}$ sequences are known to the researcher. First, consider the forecasts from the AR(1) model $y_t = a_0 + a_1 y_{t-1} + \varepsilon_t$. Updating one period, we obtain

$$y_{t+1} = a_0 + a_1 y_t + \varepsilon_{t+1}$$

If you know the coefficients a_0 and a_1, you can forecast y_{t+1} conditional on the information available at period t as

$$E_t y_{t+1} = a_0 + a_1 y_t \tag{2.47}$$

where $E_t y_{t+j}$ is a short-hand way to write the conditional expectation of y_{t+j} given the information available at t. Formally, $E_t y_{t+j} = E(y_{t+j} | y_t, y_{t-1}, y_{t-2}, \ldots, \varepsilon_t, \varepsilon_{t-1}, \ldots)$.

In the same way, since $y_{t+2} = a_0 + a_1 y_{t+1} + \varepsilon_{t+2}$, the forecast of y_{t+2} conditioned on the information available at period t is

$$E_t y_{t+2} = a_0 + a_1 E_t y_{t+1}$$

and using (2.47)

$$E_t y_{t+2} = a_0 + a_1 (a_0 + a_1 y_t)$$

Thus, the forecast of y_{t+1} can be used to forecast y_{t+2}. The point is that forecasts can be constructed using forward iteration; the forecast of y_{t+j} can be used to forecast y_{t+j+1}. Since $y_{t+j+1} = a_0 + a_1 y_{t+j} + \varepsilon_{t+j+1}$, it immediately follows that

$$E_t y_{t+j+1} = a_0 + a_1 E_t y_{t+j} \tag{2.48}$$

From (2.47) and (2.48), it should be clear that it is possible to obtain the entire sequence of j-step-ahead forecasts by forward iteration. Consider

$$E_t y_{t+j} = a_0(1 + a_1 + a_1^2 + \cdots + a_1^{j-1}) + a_1^j y_t$$

This equation, called the **forecast function**, expresses all of the j-step-ahead forecasts as a function of the information set in period t. Unfortunately, the quality of the forecasts declines as we forecast further out into the future. Think of (2.48) as a first-order difference equation in the $\{E_t y_{t+j}\}$ sequence. Since $|a_1| < 1$, the difference equation is stable, and it is straightforward to find the particular solution to the difference equation. If we take the limit of $E_t y_{t+j}$ as $j \to \infty$, we find that $E_t y_{t+j} \to a_0/(1 - a_1)$. This result is really quite general: *For any stationary ARMA model, the conditional forecast of y_{t+j} converges to the unconditional mean as $j \to \infty$.*

Because the forecasts from an ARMA model will not be perfectly accurate, it is important to consider the properties of the forecast errors. Forecasting from time period t, we can define the j-step-ahead forecast error, called $e_t(j)$, as the difference between the realized value of y_{t+j} and the forecasted value:

$$e_t(j) \equiv y_{t+j} - E_t y_{t+j}$$

Since the one-step-ahead forecast error is equivalent to $e_t(1) = y_{t+1} - E_t y_{t+1} = \varepsilon_{t+1}$, $e_t(1)$ is precisely the "unforecastable" portion of y_{t+1}, given the information available in t.

To find the two-step-ahead forecast error, we need to form $e_t(2) = y_{t+2} - E_t y_{t+2}$. Since $y_{t+2} = a_0 + a_1 y_{t+1} + \varepsilon_{t+2}$ and $E_t y_{t+2} = a_0 + a_1 E_t y_{t+1}$, it follows that

$$e_t(2) = a_1(y_{t+1} - E_t y_{t+1}) + \varepsilon_{t+2} = \varepsilon_{t+2} + a_1 \varepsilon_{t+1}$$

You should take a few moments to demonstrate that, for the AR(1) model, the j-step-ahead forecast error is given by

$$e_t(j) = \varepsilon_{t+j} + a_1 \varepsilon_{t+j-1} + a_1^2 \varepsilon_{t+j-2} + a_1^3 \varepsilon_{t+j-3} + \cdots + a_1^{j-1} \varepsilon_{t+1} \tag{2.49}$$

Since the mean of (2.49) is zero, the forecasts are unbiased estimates of each value y_{t+j}. The proof is trivial. Since $E_t \varepsilon_{t+j} = E_t \varepsilon_{t+j-1} = \cdots = E_t \varepsilon_{t+1} = 0$, the conditional expectation of (2.49) is $E_t e_t(j) = 0$. Since the expected value of the forecast error is zero, the forecasts are unbiased.

Although unbiased, the forecasts from an ARMA model are necessarily inaccurate. To find the variance of the forecast error, continue to assume that the elements of the $\{\varepsilon_t\}$ sequence are independent with a variance equal to σ^2. Hence, from (2.49), the variance of the forecast error is

$$\mathrm{var}[e_t(j)] = \sigma^2[1 + a_1^2 + a_1^4 + a_1^6 + \cdots + a_1^{2(j-1)}] \tag{2.50}$$

Thus, the one-step-ahead forecast error variance is σ^2, the two-step-ahead forecast error variance is $\sigma^2(1 + a_1^2)$, and so forth. The essential point to note is that the variance of the forecast error is an increasing function of j. As such, you can have more confidence in short-term forecasts than in long-term forecasts. In the limit as $j \to \infty$, the forecast error variance converges to $\sigma^2/(1 - a_1{}^2)$; hence, the forecast error variance converges to the unconditional variance of the $\{y_t\}$ sequence.

Moreover, assuming that the $\{\varepsilon_t\}$ sequence is normally distributed, you can place confidence intervals around the forecasts. The one-step-ahead forecast of y_{t+1} is $a_0 + a_1 y_t$, and the forecast error is σ^2. As such, the 95% confidence interval for the one-step-ahead forecast can be constructed as

$$a_0 + a_1 y_t \pm 1.96\sigma$$

We can construct a confidence interval for the two-step-ahead forecast error in the same way. From (2.48), the two-step-ahead forecast is $a_0(1 + a_1) + a_1^2 y_t$, and (2.50) indicates that var$[e_t(2)]$ is $\sigma^2(1 + a_1^2)$. Thus, the 95% confidence interval for the two-step-ahead forecast is

$$a_0(1 + a_1) + a_1^2 y_t \pm 1.96\sigma(1 + a_1^2)^{1/2}$$

Higher-Order Models

To generalize the discussion, it is possible to use the iterative technique to derive the forecasts for any ARMA(p, q) model. To keep the algebra simple, consider the ARMA(2, 1) model

$$y_t = a_0 + a_1 y_{t-1} + a_2 y_{t-2} + \varepsilon_t + \beta_1 \varepsilon_{t-1} \tag{2.51}$$

Updating one period yields

$$y_{t+1} = a_0 + a_1 y_t + a_2 y_{t-1} + \varepsilon_{t+1} + \beta_1 \varepsilon_t$$

If we continue to assume that (1) all coefficients are known; (2) all variables subscripted $t, t-1, t-2, \ldots$ are known at period t; and (3) $E_t \varepsilon_{t+j} = 0$ for $j > 0$, the conditional expectation of y_{t+1} is

$$E_t y_{t+1} = a_0 + a_1 y_t + a_2 y_{t-1} + \beta_1 \varepsilon_t \tag{2.52}$$

Equation (2.52) is the one-step-ahead forecast of y_{t+1}. The one-step-ahead forecast error is the difference between y_{t+1} and $E_t y_{t+1}$ so that $e_t(1) = \varepsilon_{t+1}$. To find the two-step-ahead forecast, update (2.51) by two periods:

$$y_{t+2} = a_0 + a_1 y_{t+1} + a_2 y_t + \varepsilon_{t+2} + \beta_1 \varepsilon_{t+1}$$

The conditional expectation of y_{t+2} is

$$E_t y_{t+2} = a_0 + a_1 E_t y_{t+1} + a_2 y_t \tag{2.53}$$

Equation (2.53) expresses the two-step-ahead forecast in terms of the one-step-ahead forecast and current value of y_t. Combining (2.52) and (2.53) yields

$$\begin{aligned} E_t y_{t+2} &= a_0 + a_1[a_0 + a_1 y_t + a_2 y_{t-1} + \beta_1 \varepsilon_t] + a_2 y_t \\ &= a_0(1 + a_1) + [a_1^2 + a_2] y_t + a_1 a_2 y_{t-1} + a_1 \beta_1 \varepsilon_t \end{aligned}$$

To find the two-step-ahead forecast error, subtract (2.53) from y_{t+2}. Thus,

$$e_t(2) = a_1(y_{t+1} - E_t y_{t+1}) + \varepsilon_{t+2} + \beta_1 \varepsilon_{t+1} \tag{2.54}$$

Since $y_{t+1} - E_t y_{t+1}$ is the one-step-ahead forecast error, we can write the forecast error as

$$e_t(2) = (a_1 + \beta_1)\varepsilon_{t+1} + \varepsilon_{t+2} \tag{2.55}$$

Finally, all the j-step-ahead forecasts can be obtained from

$$E_t y_{t+j} = a_0 + a_1 E_t y_{t+j-1} + a_2 E_t y_{t+j-2}, \ \ j \geq 2 \tag{2.56}$$

Equation (2.56) demonstrates that the forecasts will satisfy a second-order difference equation. As long as the characteristic roots of (2.56) lie inside the unit circle, the forecasts will converge to the unconditional mean: $a_0/(1 - a_1 - a_2)$. We can use (2.56) to find the j-step-ahead forecast errors. Since $y_{t+j} = a_0 + a_1 y_{t+j-1} + a_2 y_{t+j-2} + \varepsilon_{t+j} + \beta_1 \varepsilon_{t+j-1}$, the j-step-ahead forecast error is

$$\begin{aligned} e_t(j) &= a_1(y_{t+j-1} - E_t y_{t+j-1}) + a_2(y_{t+j-2} - E_t y_{t+j-2}) + \varepsilon_{t+j} + \beta_1 \varepsilon_{t+j-1} \\ &= a_1 e_t(j-1) + a_2 e_t(j-2) + \varepsilon_{t+j} + \beta_1 \varepsilon_{t+j-1} \end{aligned}$$

It should be clear that forecasts from any stationary ARMA(p, q) process will eventually satisfy the pth order difference equation comprising the homogeneous portion of the model. As such, the multistep-ahead forecasts will converge to the long-run mean of the series.

Forecast Evaluation

Now that you have estimated a series and have forecasted its future values, the obvious question is, "How good are my forecasts?" Typically, there will be several plausible models that you can select to use for your forecasts. Do not be fooled into thinking that the one with the best fit is the one that will forecast the best. To make a simple point, suppose you wanted to forecast the future values of the ARMA(2, 1) process given by (2.51). If you could forecast the value of y_{T+1} using (2.52), you would obtain the one-step-ahead forecast error

$$e_T(1) = y_{T+1} - a_0 - a_1 y_T - a_2 y_{T-1} - \beta_1 \varepsilon_T = \varepsilon_{T+1} \tag{2.57}$$

Since the forecast error is the pure *unforecastable* portion of y_{T+1}, no other ARMA model can provide you with superior forecasting performance. As such, it appears that the "true" model will provide superior forecasts to those from any other possible model. In practice, you will not know the actual order of the ARMA process or the actual values of the coefficients of that process. Instead, to create out-of-sample forecasts, it is necessary to use the estimated coefficients from what you believe to be the most appropriate form of an ARMA model. Let a hat or caret (i.e.: ^) over a parameter denote the estimated value of a parameter, and let $\{\hat{\varepsilon}_t\}$ denote the residuals of the estimated model. Hence, if you use the estimated model, the one-step-ahead forecast will be

$$E_T y_{T+1} = \hat{a}_0 + \hat{a}_1 y_T + \hat{a}_2 y_{T-1} + \hat{\beta}_1 \hat{\varepsilon}_T \tag{2.58}$$

and the one-step-ahead forecast error will be

$$e_T(1) = y_{T+1} - (\hat{a}_0 + \hat{a}_1 y_T + \hat{a}_2 y_{T-1} + \hat{\beta}_1 \hat{\varepsilon}_T)$$

Clearly, this forecast will not be identical to that from (2.57). When we forecast using (2.58), the coefficients (and the residuals) are estimated imprecisely. The forecasts made using the estimated model extrapolate this coefficient uncertainty into the future. Since coefficient uncertainty increases as the model becomes more complex, it could be that an estimated AR(1) model forecasts the process given by (2.51) better than an estimated ARMA(2, 1) model. The general point is that large models usually contain in-sample estimation errors that induce forecast errors. As shown in the studies by Clark and West (2007), Dimitrios and Guerard (2004), and Liu and Enders (2003), forecasts using overly parsimonious models with little parameter uncertainty can provide better forecasts than models consistent with the actual data-generating process. Moreover, it is very difficult to construct confidence intervals for this type of forecast error. Not only is it necessary to include the effects of the stochastic variation in the future values of $\{y_{T+i}\}$, but also it is necessary to incorporate the fact that the coefficients are estimated with error.

How do you know which one of the several reasonable models has the best forecasting performance? One way to answer this question is to put the alternative models to a head-to-head test. Since the future values of the series are unknown, you can hold back a portion of the observations from the estimation process. As such, you can estimate the alternative models over the shortened span of data and use these estimates to forecast the observations of the holdback period. You can then compare the properties of the forecast errors from the two models. To take a simple example, suppose that $\{y_t\}$ contains a total of 150 observations and that you are unsure as to whether an AR(1) or an MA(1) model best captures the behavior of the series.

One way to proceed is to use the first 100 observations to estimate both models and use each to forecast the value of y_{101}. Since you know the actual value of y_{101}, you can construct the forecast error obtained from the AR(1) and from the MA(1). These two forecast errors are precisely those that someone would have made if they had been making a one-step-ahead forecast in period 100. Now, reestimate an AR(1) and an MA(1) model using the first 101 observations. Although the estimated coefficients will change somewhat, they are those that someone would have obtained in period 101. Use the two models to forecast the value of y_{102}. Given that you know the actual value of y_{102}, you can construct two more forecast errors. Since you know all values of the $\{y_t\}$ sequence through period 150, you can continue this process so as to obtain two series of one-step-ahead forecast errors, each containing 50 observations. To keep the notation simple, let $\{f_{1i}\}$ and $\{f_{2i}\}$ denote the sequence of forecasts from the AR(1) and the MA(1), respectively. If you understand the notation, it should be clear that $f_{11} = E_{100}y_{101}$ is the first forecast using the AR(1) and $f_{2,50}$ is the last forecast from the MA(1).

Obviously, it is desirable that the forecast errors have a mean near zero and a small variance. A regression-based method to assess the forecasts is to use the 50 forecasts from the AR(1) to estimate an equation of the form

$$y_{100+i} = a_0 + a_1 f_{1i} + v_{1i} \qquad i = 1, \ldots, 50$$

If the forecasts are unbiased, an F-test should allow you to impose the restriction $a_0 = 0$ and $a_1 = 1$. Similarly, the residual series $\{v_{1i}\}$ should act as a white-noise process. It is a good idea to plot $\{v_{1i}\}$ to determine if there are periods in which your forecasts are especially poor. Now repeat the process with the forecasts from the MA(1). In particular, use the 50 forecasts from the MA(1) to estimate

$$y_{100+i} = b_0 + b_1 f_{2i} + v_{2i} \qquad i = 1, \ldots, 50$$

Again, if you use an F-test, you should not be able to reject the joint hypothesis $b_0 = 0$ and $b_1 = 1$. If the significance levels from the two F-tests are similar, you might select the model with the smallest residual variance; that is, select the AR(1) if $\text{var}(v_1) < \text{var}(v_2)$.[4]

More generally, you might want to have a holdback period that differs from 50 observations. If you have a large sample, it is possible to hold back as much as 50% of the data set. Also, you might want to use the j-step-ahead forecasts instead of the one-step-ahead forecasts. For example, if you have quarterly data and want to forecast 1 year into the future, you can perform the analysis using the four-step-ahead forecasts. Once you have the two sequences of forecast errors, you can compare their properties. With a very small sample, it may not be possible to hold back many observations. Small samples are a problem since Ashley (2003) showed that very large samples are often necessary to reveal a significant difference between the out-of-sample forecasting performances of similar models. You need to have enough observations to have well-estimated coefficients for the in-sample period and enough out-of-sample forecasts so that the test has good power.

Instead of focusing on the bias, many researchers would select the model with the smallest mean square prediction error (MSPE). Suppose you construct H one-step-ahead forecasts from two different models. Again, let f_{1i} be the forecasts from model 1 and f_{2i} be the forecasts from model 2. Since we are using the one-step-ahead forecasts, we can suppress the subscript j and denote the two series of forecasts errors as e_{1i} and e_{2i}. As such, the MSPE of model 1 can be calculated as

$$\text{MSPE} = \frac{1}{H}\sum_{i=1}^{H} e_{1i}^2$$

Several methods have been proposed to determine whether one MSPE is statistically different from the other. If you put the larger of the two MSPEs in the numerator, a standard recommendation is to use the F-statistic

$$F = \sum_{i=1}^{H} e_{1i}^2 \Big/ \sum_{i=1}^{H} e_{2i}^2 \tag{2.59}$$

The intuition is that the value of F will equal unity if the forecast errors from the two models are identical. A very large value of F implies that the forecast errors from the first model are substantially larger than those from the second. Under the null hypothesis of equal forecasting performance, (2.59) has a standard F-distribution with (H, H) degrees of freedom if the following three assumptions hold:

1. The forecast errors have zero mean and are normally distributed.
2. The forecast errors are serially uncorrelated.
3. The forecast errors are contemporaneously uncorrelated with each other.

Although it is common practice to assume that the $\{\varepsilon_t\}$ sequence is normally distributed, it is not necessarily the case that the forecast errors are normally distributed with a mean value of zero. Similarly, the forecasts may be serially correlated; this is particularly true if you use multistep-ahead forecasts. For example, equation (2.55) indicated that the two-step-ahead forecast error for y_{t+2} is

$$e_t(2) = (a_1 + \beta_1)\varepsilon_{t+1} + \varepsilon_{t+2}$$

and updating by one period yields the two-step-ahead forecast error for y_{t+3}:

$$e_{t+1}(2) = (a_1 + \beta_1)\varepsilon_{t+2} + \varepsilon_{t+3}$$

It should be clear that the two forecast errors are correlated. In particular,

$$E[e_t(2)e_{t+1}(2)] = (a_1 + \beta_1)\sigma^2$$

The point is that predicting y_{t+2} from the perspective of period t and predicting y_{t+3} from the perspective of period $t + 1$ both contain an error due to the presence of ε_{t+2}. However, for $i > 1$, $E[e_t(2)e_{t+i}(2)] = 0$ since there are no overlapping forecasts. Hence, the autocorrelations of the two-step-ahead forecast errors cut to zero after lag 1. You should be able to demonstrate the general result that the j-step-ahead forecast errors act as an MA($j - 1$) process.

Finally, the forecast errors from the two alternative models will usually be highly correlated with each other. For example, a negative realization of ε_{t+1} will tend to cause the forecasts from both models to be too high. Unfortunately, the violation of any one of these assumptions means that the ratio of the MSPEs in (2.59) does not have an F-distribution.

THE GRANGER–NEWBOLD TEST Granger and Newbold (1976) show how to overcome the problem of contemporaneously correlated forecast errors. If you have H one-step-ahead forecast errors from each model, use the two sequences of forecast errors to form

$$x_i = e_{1i} + e_{2i} \quad \text{and} \quad z_i = e_{1i} - e_{2i} \quad i = 1, \dots, H.$$

Given that the first two assumptions above are valid, under the null hypothesis of equal forecast accuracy, x_i and z_i should be uncorrelated. Consider:

$$\rho_{xz} = Ex_iz_i = E(e_{1i}^2 - e_{2i}^2)$$

If the models forecast equally well, it follows that $Ee_{1i}^2 = Ee_{2i}^2$. Model 1 has a larger MSPE if ρ_{xz} is positive, and model 2 has a larger MSPE if ρ_{xz} is negative. Let r_{xz} denote the sample correlation coefficient between $\{x_i\}$ and $\{z_i\}$. Granger and Newbold (1976) show that if assumptions 1 and 2 hold

$$r_{xz}/\sqrt{(1 - r_{xz}^2)/(H - 1)} \tag{2.60}$$

has a t-distribution with $H - 1$ degrees of freedom. Thus, if r_{xz} is statistically different from zero, model 1 has a larger MSPE if r_{xz} is positive, and model 2 has a larger MSPE if r_{xz} is negative.

THE DIEBOLD–MARIANO TEST There is a very large literature trying to extend the Granger–Newbold test so as to relax assumptions 1 and 2. Moreover, applied econometricians might be interested in measures of forecasting performance other than the sum of squared errors. Indeed, it should be clear that using the sum of squared errors as a criterion makes sense only if the loss from making an incorrect forecast is quadratic. However, there are many other possibilities. For example, if your loss depends on the size of the forecast error, you should be concerned with the absolute values of the forecast errors. Alternatively, an options trader receives a payoff of zero if the value of the underlying asset lies below the strike price but receives a one-dollar payoff for each dollar the asset price rises above the strike price. In such a circumstance, the loss payoff is asymmetric. Diebold and Mariano (1995) have developed a test that relaxes assumptions 1–3 and allows for an objective function that is not quadratic.

As before, if we consider only one-step-ahead forecasts, we can eliminate the subscript j. As such, we can let the loss from a forecast error in period i be denoted by $g(e_i)$. In the typical case of mean-squared errors, the loss is e_i^2. Nevertheless, to allow the loss function to be general, we can write the differential loss in period i from using model 1 versus model 2 as $d_i = g(e_{1i}) - g(e_{2i})$. The mean loss can be obtained as

$$\bar{d} = \frac{1}{H}\sum_{i=1}^{H}[g(e_{1i}) - g(e_{2i})] \tag{2.61}$$

Under the null hypothesis of equal forecast accuracy, the value of $\bar{d}$ is zero. Since $\bar{d}$ is the mean of the individual losses, under fairly weak conditions, the central limit theorem implies that $\bar{d}$ should have a normal distribution. Hence, it is not necessary to assume that the individual forecast errors are normally distributed. Thus, if we knew $\text{var}(\bar{d})$, we could construct the ratio $\bar{d}/\sqrt{\text{var}(\bar{d})}$ and test the null hypothesis of equal forecast accuracy using a standard normal distribution. In practice, the implementation of the test is complicated by the fact that we need to estimate $\text{var}(\bar{d})$.

If the $\{d_i\}$ series is serially uncorrelated with a sample variance of γ_0, the estimate of $\text{var}(\bar{d})$ is simply $\gamma_0/(H - 1)$. Since we use the estimated value of the variance, the expression $\bar{d}/\sqrt{\gamma_0/(H-1)}$ has a t-distribution with $H - 1$ degrees of freedom.

There is a very large literature on the best way to estimate the standard deviation of $\bar{d}$ in the presence of serial correlation. Many of the technical details are not appropriate here. Diebold and Mariano let γ_i denote the ith autocovariance of the d_t sequence. Suppose that the first q values of γ_i are different from zero. The variance of $\bar{d}$ can be approximated by $\text{var}(\bar{d}) = [\gamma_0 + 2\gamma_1 + \cdots + 2\gamma_q](H-1)^{-1}$; the standard deviation is the square root. As such, Harvey, Leybourne, and Newbold (1998) recommended constructing the Diebold–Mariano (DM) statistic as

$$\text{DM} = \bar{d}/\sqrt{(\gamma_0 + 2\gamma_1 + \cdots + 2\gamma_q)/(H-1)} \tag{2.62}$$

Compare the sample value of (2.62) to a t-statistic with $H-1$ degrees of freedom. As a practical matter, a simple way to proceed is to regress the d_i on a constant and use a t-test (with robust standard errors) to determine whether the constant is statistically different from zero. Note that the construction of (2.62) can be sensitive to the choice of q, and if one or more of the $\gamma_i < 0$, the estimated variance can be negative. In such circumstances, it is preferable to use robust standard errors—such as those in Newey and West (1987). All professional software packages allow you to directly obtain the Newey–West estimator of the variance. Additional details are included in the *Supplementary Manual*.

It is also possible to use the method for the j-step-ahead forecasts $e_{1i}(j)$ and $e_{2i}(j)$. Construct each $d_i = g(e_{1i}(j)) - g(e_{2i})$ and the mean $\bar{d}$. If you construct H forecast errors, the DM statistic is

$$\mathrm{DM} = \bar{d}/\sqrt{(\gamma_0 + 2\gamma_1 + \cdots + 2\gamma_q)/[H + 1 - 2j + H^{-1}j(j-1)]}.$$

An example showing the appropriate use of the Granger–Newbold and Diebold–Mariano tests is provided in Section 10. Nevertheless, before proceeding, a strong word of caution is in order. Clark and McCracken (2001) show that the Granger–Newbold and Diebold–Mariano tests have a t-distribution only when the underlying forecasting models are not nested. For example, the tests might not work well when comparing forecasts from an AR(1) model to those obtained from an ARMA(2, 1) model. Clearly, the AR(1) can be obtained from the ARMA(2, 1) specification by setting $a_2 = \beta_1 = 0$. The problem with nested models is that under the null hypothesis of equal MSPEs (so that the data are generated by the small model), the two models should predict equally well. However, the large model will always contain some extra error as it contains unnecessary parameters. Hence, if you want to test whether the data are actually generated from the different models, you need to control for the parameter uncertainty.

Clark and West (2007) develop a simple procedure to adjust the forecast errors from the large model so that a simple variant of the DM statistic can be used with nested models. To continue with the notation developed above, denote the H forecasts from model 1 as f_{1i} and the forecast errors as e_{1i}. Similarly, the H forecasts and forecast errors from model 2 are f_{2i} and e_{2i}, respectively. Let model 1 be nested within model 2. Given that the models are nested, the sole reason for any discrepancy between f_{1i} and f_{2i} is due to parameter estimation error. If this estimation error is subtracted from e_{2i}, the adjusted forecast errors can be used as the basis for the modified DM test. Consider the z_i series constructed from the squares of these errors as

$$z_i = (e_{1i})^2 - \left[\left(e_{2i}\right)^2 - (f_{1i} - f_{2i})^2\right] \qquad i = 1, \ldots, H.$$

Allowing for parameter uncertainty, under the null hypothesis of that the two models predict equally well, z_i should be zero. Under the alternative hypothesis, the data are generated from model 2. Hence, to perform the test, regress the z_i series on a constant. Since the test is one sided, if the t-statistic for the constant exceeds 1.645, reject the null hypothesis of equal forecast accuracy at the 5% significance level. If you reject the null hypothesis, conclude that the data are generated from model 2. Otherwise, the data are more likely to be generated from model 1. If the $\{z_i\}$ series is serially correlated, you should perform the test with a robust t-statistic, such as that in the study by Newey and West (1987).

10. A MODEL OF THE INTEREST RATE SPREAD

The term "textbook example" is supposed to connote a very clear-cut illustration. If you are looking for a textbook example of the Box–Jenkins methodology, go back to Section 7 or turn to Question 11 at the end of this chapter. In practice, we rarely find a data series that precisely conforms to a theoretical ACF or PACF. This section is intended to illustrate some of the ambiguities that can be encountered when using the Box–Jenkins technique. These ambiguities may lead two equally skilled econometricians to estimate and forecast the same series using very different ARMA processes. Many view the necessity of relying on the researcher's judgment and experience as a serious weakness of a procedure that is designed to be scientific. Yet, if you make reasonable choices, you will select models that come very close to mimicking the actual data-generating process.

It is useful to illustrate the Box–Jenkins modeling procedure by estimating a quarterly model of the spread between a long-term and a short-term interest rate. Specifically, the interest rate spread (s_t) can be formed as the difference between the interest rate on 5-year U.S. government bonds and the rate on 3-month treasury bills. The data used in this section are the series labeled R5 and TBILL in the file QUARTERLY.XLS. Exercise 12 at the end of this chapter will help you to reproduce the results reported below.

Panel (a) of Figure 2.5 shows the spread over the period from 1960Q1 to 2012Q4. Although there are a few instances in which the spread is negative, the difference between long- and short-term rates is generally positive (the sample mean is 1.21). Notice that the series shows a fair amount of persistence in that the durations when the spread is above or below the mean can be quite lengthy. Moreover, there do not appear to be any major structural breaks (such as a permanent jump in the mean or variance) in that the dynamic nature of the process seems to be constant over time. As such, it is quite reasonable to suppose that the $\{s_t\}$ sequence is covariance stationary. In contrast, as shown in Panel (b), the first difference of the spread seems to be very erratic. As you will verify in Exercise 12, the Δs_t series has little informational content that can be used to forecast its future values. As such, it seems reasonable to estimate a model of the $\{s_t\}$ sequence without any further transformations. Nevertheless, because there are several large positive and negative jumps in the value of s_t, some researchers might want to transform it so as to diminish its volatility. A reasonable number of such shocks might indicate a departure from the assumption that the errors are normally distributed. Although a logarithmic or a square root transformation is impossible because some realizations of s_t are negative, one could dampen the series using $y_t = \log(s_t + 3)$. The point is that you should always maintain a healthy skepticism of the accuracy of your model since the behavior of the data-generating process may not fully conform to the underlying assumptions of the methodology.

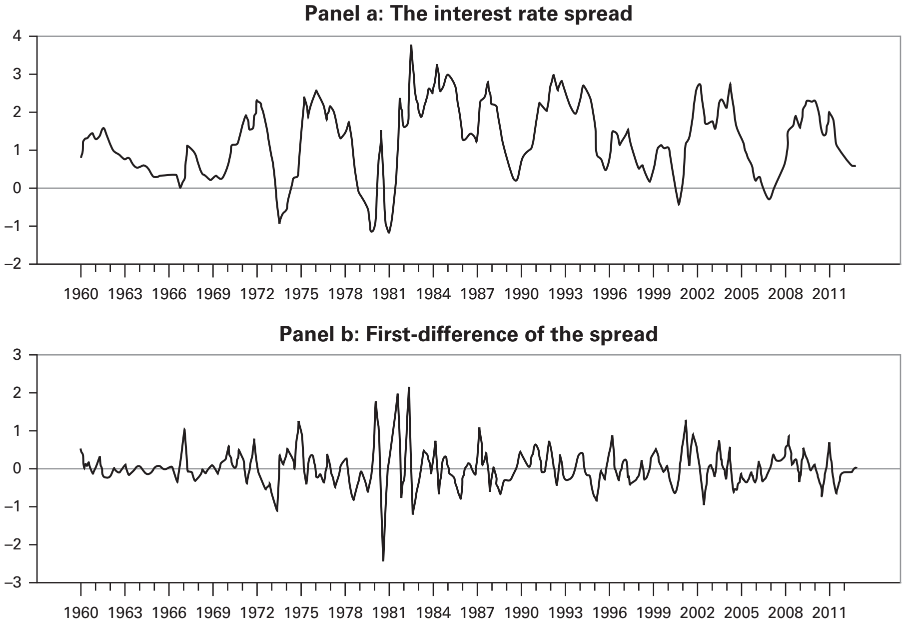


FIGURE 2.5 Time Path of the Interest Rate Spread

Before reading on, you should examine the autocorrelations and partial autocorrelation functions of the $\{s_t\}$ sequence shown in Figure 2.6. Try to identify the tentative models that you would want to estimate. Recall that the theoretical ACF of a pure MA(q) process cuts off to zero at lag q, and the theoretical ACF of an AR(1) model decays geometrically. Examination of Figure 2.6 suggests that neither of these

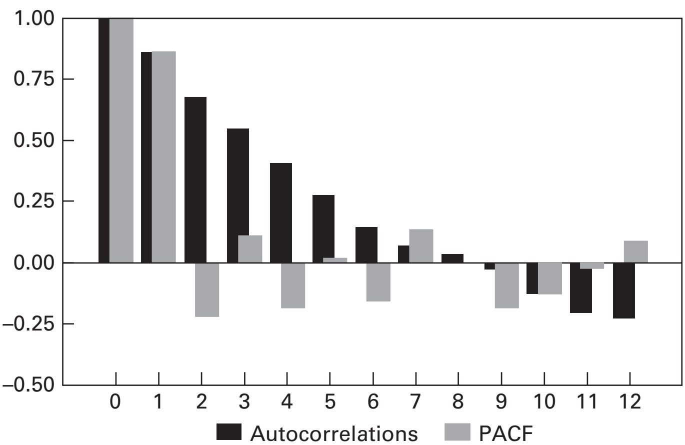


FIGURE 2.6 ACF and PACF of the Spread

specifications perfectly describes the sample data. In selecting your set of plausible models, also note the following:

1. The ACF and PACF converge to zero quickly enough that we do not have to worry about a time-varying mean. As suggested above, we do not want to *overdifference* the data and try to model the $\{\Delta s_t\}$ sequence.
2. The ACF does not cut to zero so that we can rule out a pure MA(q) process.
3. The ACF is not really suggestive of a pure AR(1) process in that the decay does not appear to be geometric. The value of ρ_1 is 0.857, and the values of ρ_2, ρ_3, and ρ_4 are 0.678, 0.550, and 0.411, respectively.
4. The estimated values of the PACF are such that $\phi_{11} = 0.858$, $\phi_{22} = -0.217$, $\phi_{33} = 0.112$, and $\phi_{44} = -0.188$. Although ϕ_{55} is close to zero, $\phi_{66} = -0.151$ and $\phi_{77} = 0.136$. Recall that, under the null hypothesis of a pure AR(p) model, the variance of $\phi_{p+i,p+i}$ is approximately equal to $1/T$. Since there are 212 total observations, the values of ϕ_{22}, ϕ_{44}, and ϕ_{66} are more than two standard deviations from zero (i.e., $2/212^{0.5} = 0.138$). In a pure AR(p) model, the PACF cuts to zero after lag p. Hence, if the s_t series follows a pure AR(p) process, the value of p could be as high as six or seven.
5. There appears to be an oscillating pattern in the PACF in that the first seven values alternate in sign. Oscillating decay of the PACF is characteristic of a positive MA coefficient.

Due to the number of small and marginally significant coefficients, the ACF and PACF of the spread are probably more ambiguous than most of those you will encounter. Hence, suppose you do not know where to start and estimate the s_t series using a pure AR(p) model. To illustrate the point, if you estimate the s_t series as an AR(7) process, you should obtain the estimates given in column 2 of Table 2.4. If you examine the table, you will find that all of the t-statistics on the first six lags exceed 1.96 in absolute value (indicating that the coefficients are significant the 5% level). Since t-statistic on the coefficient for y_{t-7} is 1.93, it is unclear as to whether to include the seventh lag. The sum or squared residuals (SSR) is 43.86 and the AIC and SBC are 791.10 and 817.68, respectively. The significance levels of the Q-statistics for lags 4, 8, and 12 indicate no remaining autocorrelation in the residuals.

Although the AR(7) model has some desirable attributes, one reasonable estimation strategy is to eliminate the seventh lag and estimate an AR(6) model over the same sample period. [Note that the data set begins in 1960Q1, so that with seven lags the estimation of the AR(7) begins in 1961Q4.] Although the autocorrelations of the residuals are such that $\rho_8 = 0.20$, the significance levels of the Q(4), Q(8), and Q(12) statistics (equal to 0.29, 10.93, and 16.75) are 0.99, 0.21, and 0.16, respectively. As such, the Q-statistics suggest that you should not try to account to account for the residual autocorrelations at lag 8. Although a_5 appears to be statistically insignificant, it is generally not a good idea to use t-statistics to eliminate intermediate lags. As such, most researchers would not eliminate the fifth lag and estimate a model with lags 1 through 4 and lag 6. Recall that the appropriate use of a t-statistic requires that regressor in

Table 2.4 Estimates of the Interest Rate Spread

	AR (7)	AR (6)	AR (2)	$p = 1,2,7$	ARMA (1,1)	ARMA (2,1)	$p = 2$; $ma = (1,7)$
a_0	1.20	1.20	1.19	1.19	1.19	1.19	1.20
	(6.57)	(7.55)	(6.02)	(6.80)	(6.16)	(5.56)	(5.74)
a_1	1.11	1.09	1.05	1.04	0.76	0.43	0.36
	(15.76)	(15.54)	(15.25)	(14.83)	(14.69)	(2.78)	(3.15)
a_2	−0.45	−0.43	−0.22	−0.20		0.31	0.38
	(−4.33)	(−4.11)	(−3.18)	(−2.80)		(2.19)	(3.52)
a_3	0.40	0.36					
	(3.68)	(3.39)					
a_4	−0.30	−0.25					
	(−2.70)	(−2.30)					
a_5	0.22	0.16					
	(2.02)	(1.53)					
a_6	−0.30	−0.15					
	(−2.86)	(−2.11)					
a_7	0.14			−0.03			
	(1.93)			(−0.77)			
β_1					0.38	0.69	0.77
					(5.23)	(5.65)	(9.62)
β_7							−0.14
							(−3.27)
SSR	43.86	44.68	48.02	47.87	46.93	45.76	43.72
AIC	791.10	792.92	799.67	801.06	794.96	791.81	784.46
SBC	817.68	816.18	809.63	814.35	804.93	805.10	801.07
$Q(4)$	0.18	0.29	8.99	8.56	6.63	1.18	0.76
$Q(8)$	5.69	10.93	21.74	22.39	18.48	12.27	2.60
$Q(12)$	13.67	16.75	29.37	29.16	24.38	19.14	11.13

Notes:
To ensure comparability, each equation was estimated over the 1961Q4 – 2012Q4 period.
Values in parentheses are the t-statistics for the null hypothesis that the estimated coefficient is equal to zero. SSR is the sum of squared residuals. $Q(n)$ are the Ljung–Box Q-statistics of the residual autocorrelations.
For ARMA models, many software packages do not actually report the intercept term a_0. Instead, they report the estimated mean of process, μ_y along with the t-statistic for the null hypothesis that $\mu_y = 0$. The historical reason for this convention is that it was easier to first demean the data and then estimate the ARMA coefficients than to estimate all values in one step. If your software package reports a constant term approximately equal to 0.216, it is reporting the estimated intercept.

question be uncorrelated with the other regressors. Given the autoregressive nature of the series, y_{t-5} is certainly correlated with y_{t-4} and y_{t-6}. The overall result is that the diagnostic checks of the AR(6) model suggest that it is adequate. In comparing the AR(6) and AR(7) models, the AIC selects the AR(7) model, whereas the SBC selects the more parsimonious AR(6) model.

Suppose that you try a very parsimonious model and estimate an AR(2). As you can see from the fourth column of the table, the AIC selects the AR(7) model, but SBC selects the AR(2) model. However, the residual autocorrelations from the AR(2) are problematic in that

ρ_1	ρ_2	ρ_3	ρ_4	ρ_5	ρ_6	ρ_7	ρ_8
0.03	−0.13	0.16	0.01	0.08	−0.10	−0.14	0.16

The Q-statistics from the AR(2) model indicate significant autocorrelation in the residuals at the shorter lags. As such, it should be eliminated from further consideration.

If you examined the AR(7) carefully, you might have noticed that a_3 almost offsets a_4 and that a_5 almost offsets a_6 (since $a_3 + a_4 \approx 0$ and $a_5 + a_6 \approx 0$). If you reestimate the model without s_{t-3}, s_{t-4}, s_{t-5}, and s_{t-6}, you should obtain the results given in column 5 of Table 2.4. Since the coefficient for s_{t-7} is now statistically insignificant, it might seem preferable to use the AR(2) instead. Yet, the AR(2) has been shown to be inadequate relative to the AR(7) and the AR(6) models.

Even though the AR(6) and AR(7) models perform relatively well, they are not necessarily the best forecasting models. There are several possible alternatives since the patterns of the ACF and PACF are not immediately clear. Results for a number of models with MA terms are shown in columns 6, 7, and 8 of Table 2.4:

1. From the decaying ACF, someone might try to estimate the ARMA(1, 1) model reported in column 6 of the table. The estimated value of a_1 (0.76) is statistically different from zero and is almost five standard deviations from unity. The estimated value of β_1 (0.38) is statistically different from zero and implies that the process is invertible. Notice that the SBC from the ARMA(1, 1) is smaller than that of the AR(7) and the AR(6). Nevertheless, the ARMA(1, 1) specification is inadequate because of remaining serial correlation in the residuals. The Ljung–Box Q-statistic for four lags of the residuals (equal to 6.63) has a significance level of 15.7%. As such, we cannot reject the null that $Q(4) = 0$ any conventional significance level. However, the $Q(8)$ and $Q(12)$ statistics indicate that the residuals from this model exhibit substantial serial autocorrelation. As such, we must eliminate the ARMA(1, 1) model from consideration.
2. Since the ACF decays and the PACF seems to oscillate beginning with lag 2 ($\phi_{22} = -0.217$), it seems plausible to estimate an ARMA(2, 1) model. As shown in column 6 of the table, the model is an improvement over the ARMA(1, 1) specification. The estimated coefficients ($a_1 = 0.43$ and $a_2 = 0.31$) are each significantly different from zero at conventional levels and imply characteristic roots in the unit circle. The AIC selects the ARMA(2, 1) model over that AR(6) and the SBC selects the ARMA(2, 1) over the AR(6) and the AR(7). The values for $Q(4)$, $Q(8)$, and $Q(12)$ indicate that the autocorrelations of the residuals are not statistically significant at the 5% level. Consider the ACF of the residuals:

ρ_1	ρ_2	ρ_3	ρ_4	ρ_5	ρ_6	ρ_7	ρ_8
0.01	0.01	−0.07	−0.02	−0.03	−0.08	−0.15	0.15

3. In order to account for the serial correlation at lag 7, it might seem plausible to add an MA term to the model at lag 7. As given in the last column of the table, all of the estimated coefficients are of high quality. In particular, the coefficient for β_7 has a t-statistic of -3.27. The estimated values of a_1 and a_2 are similar to those of the ARMA(2, 1) model. Again, the Q-statistics indicate that the autocorrelations of the residuals are not significant at conventional level. Both the AIC and SBC select the ARMA[2,(1,7)] specification over any of the other models. You can easily verify that the MA coefficient at lag 7 provides a better fit than an AR coefficient at lag 7 and that an ARMA[2,(1,8)] model is inadequate.

Although the ARMA[2,(1,7)] model appears to be quite reasonable, other researchers might have selected a decidedly different model. Consider some of the alternatives listed below.

1. **Parsimony versus Overfitting:** In Section 7, we examined the issue of fitting an MA coefficient at lag 16 to a true AR(2) process. If you reexamine the example, you can understand why some researchers shy away from estimating a model with long lags lengths that are disjoint from those of other periods. In the example of the spread, the problem with the ARMA(2, 1) model is that there was a small amount of residual autocorrelation around lag 7 or 8. The addition of the MA coefficient at lag 7 yielded a model with a better fit and remedied the serial correlation problem. However, is it really plausible that ε_{t-7} has a direct effect on the current value of the interest rate spread while lags 3, 4, 5, and 6 have no direct effects? In other words, do the markets for securities work in such a way that what happens 7 quarters in the past has a larger effect on today's interest rates than events occurring in the more recent past? Moreover, as you can verify by estimating the ARMA[2,(1,7)] model, the t-statistic for β_7 over the $1982Q1-2012Q4$ period is equal to 0.60 and is not statistically significant. Notice that Panel (b) of Figure 2.5 suggests that the volatility of the spread in the late 1970s and early 1980s is not typical of the entire sample. It could be the case that the realizations from this period are anomalies that have large effects on the coefficient estimates and their standard errors. Thus, even though the AIC and SBC select the ARMA[2,(1,7)] model over the ARMA(2, 1) model, some researchers would prefer the latter.

   More generally, *overfitting* refers to a situation in which an equation is fit to some of the idiosyncrasies of present in a particular sample that are not actually representative of the data-generating process. In applied work, no data set will perfectly correspond to every assumption required for the Box–Jenkins methodology. Since it is not always clear which characteristics of the sample are actually present in the data-generating process, the attempt to expand a model so as to capture every feature of the data may lead to overfitting.

2. **Volatility:** Given the volatility of the $\{s_t\}$ series during the late 1970s and early 1980s, transforming the spread using some sort of a square root or logarithmic transformation might be appropriate. Moreover, the s_t series has a

number of sharp jumps, indicating that the assumption of normality might be violated. For a constant c such that $s_t + c$ is always positive, transformations such as $\ln(s_t + c)$ or $(s_t + c)^{0.5}$ yield series with less volatility than the s_t series itself. Alternatively, it is possible to model the difference between the log of the 5-year rate and the log of the 3-month rate.

A general class of transformations was proposed by Box and Cox (1964). Suppose that all values of $\{y_t\}$ are positive so that it is possible to construct the transformed $\{y_t^*\}$ sequence as

$$\begin{aligned} y_t^* &= (y_t^{\lambda} - 1)/\lambda \qquad && \lambda \neq 0 \\ &= \ln(y_t) && \lambda = 0 \end{aligned}$$

The common practice is to transform the data using a preselected value of λ. The selection of a value for λ that is close to zero acts to "smooth" the sequence. An ARMA model can be fitted to the transformed data. Although some software programs have the capacity to simultaneously estimate λ along with the other parameters of the ARMA model, this approach has fallen out of fashion. Instead, it is possible to actually model the variance using the methods discussed in Chapter 3.

3. **Trends:** Suppose that the span of the data had been somewhat different in that the first observation was for 1973Q1 and the last was for 2004Q4. If you examine Panel (a) of Figure 2.4, you can see that someone might be confused and believe that the data contained an upward trend. Their misinterpretation of the data might be reinforced by the fact that the ACF converges to zero rather slowly. As such, they might have estimated a model of the Δs_t series. Others might have detrended the data using a deterministic time trend.

Out-of-Sample Forecasts

We can assess the forecasting performance of the AR(7) and ARMA[2,(1,7)] models by examining their bias and mean square prediction errors. Given that the data set contains a total of 205 (i.e., $205 = 212 - 7$) usable observations, it is possible to use a holdback period of 50 observations. This way, there are at least 155 observations in each of the estimated models and an adequate number of out-of-sample forecasts. First, the two models were estimated using all available observations through 2000Q2 and the two one-step-ahead forecasts were obtained. The actual value of $s_{2000:3} = 0.40$; the AR(7) predicted a value of 0.697, and the ARMA[2,(1,7)] model predicted a value of 0.591. Thus, the forecast of the ARMA[2,(1,7)] is superior to that of the ARMA(7) for this first period. An additional 49 forecasts were obtained for periods 2000Q4 to 2012Q4. Let e_{1t} denote the forecast errors from the AR(7) model and e_{2t} denote the forecast errors from the ARMA[2,(1,7)] model. The mean of e_{1t} is 1.239, the mean of e_{2t} is 1.244, and the estimated variances are $\text{var}(e_1) = 0.797$ and $\text{var}(e_2) = 0.780$. As such, the bias of AR(7) is slightly smaller while the ARMA[2,(1,7)] has the smallest MSPE.

To ascertain whether these differences are statistically significant, we first check the bias. Let the $\{f_{1t}\}$ series contain the 50 forecasts of the AR(7) model and let $\{f_{2t}\}$

contain the 50 forecasts from the ARMA[2,(1,7)] model. Beginning with $t = 2000Q3$, we can estimate the two regression equations:

$$s_t = 0.0594 + 0.968f_{1t} \qquad \text{and} \qquad s_t = 0.004 + 1.004f_{2t}$$

For the AR(7) model, the F-statistic for the restriction that the intercept equals zero and the slope equals unity is 0.110 with significance level of 0.896. Clearly, the restriction of unbiased forecasts does not appear to be binding. For the ARMA[2,(1,7)] model, the F-statistic is 0.014 with a significance level of 0.986. Hence, there is strong evidence that both models have unbiased forecasts.

Next, consider the Granger–Newbold test for equal mean square prediction errors. Form the x_i and z_i series as $x_i = e_{1i} + e_{2i}$ and $z_i = e_{1i} - e_{2i}$, respectively. The correlation coefficient between x_i and z_i is $r_{xz} = 0.234$. Given that there are 50 observations in the holdback period, form the Granger–Newbold statistic

$$r_{xz}/\sqrt{(1 - r_{xz}^2)/(H - 1)} = 0.234/\sqrt{(1 - (0.234)^2)/49} = 1.69$$

With 49 degrees of freedom, a value of $t = 1.69$ is not statistically significant. We can conclude that the forecasting performance of the AR(7) is not statistically different from that of the ARMA[2,(1,7)].

Since the e_{1i} and e_{2i} series contain only a low amount of serial correlation, we obtain virtually the same answer using the DM statistic. Oftentimes, forecasters are concerned about the MSPE. However, there are many other possibilities. In Exercise 12 at the end of this chapter, you will be asked to use the mean absolute error. Now, to illustrate the use of the DM test, suppose that the cost of a forecast error rises extremely quickly in the size of the error. In such circumstances, the loss function might be best represented by the forecast error raised to the fourth power. Hence,

$$d_i = (e_{1i})^4 - (e_{2i})^4 \tag{2.63}$$

The mean value of the $\{d_i\}$ sequence from (2.63) (i.e., $\overline{d}$) is 0.01732, and the estimated variance is 0.002466. Since $H = 50$, we can form the DM statistic

$$\text{DM} = 0.01732/(0.002466/49)^{1/2} = 2.441$$

The null hypothesis is that the models have equal forecasting accuracy, and the alternative hypothesis is that the forecast errors from the AR[2,(1,7)] are smaller than those of the AR(7). With 49 degrees of freedom, the t-value of 2.441 is significant at the 1.829% level. Hence, there is evidence in favor of the AR[2,(1,7)] model. If there is serial correlation in the $\{d_t\}$ series, we need to use the specification in (2.63). Toward this end, we would want to select the statistically significant values of γ_q. The autocorrelations of d_t are

ρ_1	ρ_2	ρ_3	ρ_4	ρ_5	ρ_6	ρ_7	ρ_8	ρ_9	ρ_{10}	ρ_{11}	ρ_{12}
−0.10	−0.15	0.26	0.01	0.36	0.00	−0.09	0.13	0.06	0.05	−0.08	0.07

$$Q(4) = 5.53;\ Q(8) = 14.76;\ \text{and } Q(12) = 15.93$$

Although ρ_5 is large, many applied econometricians would dismiss it as spurious. It does not seem plausible that correlations for ρ_1 and ρ_2 are actually very close to zero while the correlation between d_t and d_{t-5} is very large. Moreover, the Ljung–Box $Q(4)$, $Q(8)$, and $Q(12)$ statistics do not indicate that the autocorrelations are significant. The significance levels are 0.237, 0.064, and 0.195, respectively. Nevertheless, if you do estimate the long-run variance using (2.63) with five lags, you should find that DM = 1.848 (so that the MSPEs are not statistically different from each other). The example underscores the point made earlier that there is no clear answer as to the best way to measure the long-run variance of $\bar{d}$ in the presence of serial correlation. The more general result is that the two models are not substantially different from each other. Both should provide reasonable forecasts.

11. SEASONALITY

Many economic processes exhibit some form of seasonality. The agricultural, construction, and travel sectors have obvious seasonal patterns resulting from their dependence on the weather. Similarly, the Thanksgiving-to-Christmas holiday season has a pronounced influence on the retail trade. In fact, the seasonal variation of a series may account for the preponderance of its total variance. Forecasts that ignore important seasonal patterns will have a high variance.

Too many people fall into the trap of ignoring seasonality if they are working with **deseasonalized** or **seasonally adjusted** data. Suppose you collect a data set that the U.S. Census Bureau has "seasonally adjusted" using its X–11, X–12, or X–13 methods.[5] In principle, the seasonally adjusted data should have the seasonal pattern removed. However, caution is necessary. Although a standardized procedure may be necessary for a government agency reporting hundreds of series, the procedure might not be best for an individual wanting to model a single series. Even if you use seasonally adjusted data, a seasonal pattern might remain. This is particularly true if you do not use the entire span of data; the portion of the data used in your study can display more (or less) seasonality than the overall span. There is another important reason to be concerned about seasonality when using deseasonalized data. Implicit in any method of seasonal adjustment is a two-step procedure. First, the seasonality is removed, and second, the autoregressive and moving average coefficients are estimated using Box–Jenkins techniques. As surveyed in Bell and Hillmer (1984), often the seasonal and the ARMA coefficients are best identified and estimated jointly. In such circumstances, it is wise to avoid using seasonally adjusted data.

Models of Seasonal Data

The Box–Jenkins technique for modeling seasonal data is only a bit different from that of nonseasonal data. The twist introduced by seasonal data of period s is that the seasonal coefficients of the ACF and PACF appear at lags $s, 2s, 3s, \ldots$, rather than at lags $1, 2, 3, \ldots$. For example, two purely seasonal models for quarterly data might be

$$y_t = a_4 y_{t-4} + \varepsilon_t, \qquad |a_4| < 1 \tag{2.64}$$

and

$$y_t = \varepsilon_t + \beta_4 \varepsilon_{t-4} \tag{2.65}$$

You can easily convince yourself that the theoretical correlogram for (2.64) is such that $\rho_i = (a_4)^{i/4}$ if $i/4$ is an integer and $\rho_i = 0$, otherwise; thus, the ACF exhibits decay at lags 4, 8, 12, ... For model (2.65), the ACF exhibits a single spike at lag 4, and all other correlations are zero.

In practice, identification will be complicated by the fact that the seasonal pattern will interact with the nonseasonal pattern in the data. The ACF and PACF for a combined seasonal/nonseasonal process will reflect both elements. Note that, with quarterly data, a seasonal MA term can have the form

$$y_t = a_1 y_{t-1} + \varepsilon_t + \beta_1 \varepsilon_{t-1} + \beta_4 \varepsilon_{t-4} \tag{2.66}$$

Alternatively, an autoregressive coefficient at lag 4 might have been used to capture the seasonality

$$y_t = a_1 y_{t-1} + a_4 y_{t-4} + \varepsilon_t + \beta_1 \varepsilon_{t-1}$$

Both of these methods treat the seasonal coefficients additively; an AR or an MA coefficient is added at the seasonal period. **Multiplicative seasonality** allows for the interaction of the ARMA and the seasonal effects. Consider the multiplicative specifications

$$(1 - a_1 L)y_t = (1 + \beta_1 L)(1 + \beta_4 L^4)\varepsilon_t \tag{2.67}$$

$$(1 - a_1 L)(1 - a_4 L^4)y_{\rm t} = (1 + \beta_1 L)\varepsilon_t \tag{2.68}$$

Equation (2.67) differs from (2.66) in that it allows the moving average term at lag 1 to interact with the seasonal moving average effect at lag 4. In the same way, (2.68) allows the autoregressive term at lag 1 to interact with the seasonal autoregressive effect at lag 4. Many researchers prefer the multiplicative form since a rich interaction pattern can be captured with a small number of coefficients. Rewrite (2.67) as

$$y_t = a_1 y_{t-1} + \varepsilon_t + \beta_1 \varepsilon_{t-1} + \beta_4 \varepsilon_{t-4} + \beta_1 \beta_4 \varepsilon_{t-5}$$

Estimating only three coefficients (i.e., a_1, β_1, and β_4) allows us to capture the effects of an autoregressive term and the effects of moving average terms at lags 1, 4, and 5. Of course, you do not really get something for nothing. The estimates of the three moving average coefficients are interrelated. A researcher estimating the unconstrained model $y_t = a_1 y_{t-1} + \varepsilon_t + \beta_1 \varepsilon_{t-1} + \beta_4 \varepsilon_{t-4} + \beta_5 \varepsilon_{t-5}$ would necessarily obtain a smaller residual sum of squares. However, (2.67) is clearly the more parsimonious model. If the unconstrained value of β_5 approximates the product $\beta_1 \beta_4$, the multiplicative model will be preferable. For this reason, most software packages have routines capable of estimating multiplicative models. Otherwise, there are no theoretical grounds leading us to prefer one form of seasonality over another. As illustrated in the last section, experimentation and diagnostic checks are probably the best way to obtain the most appropriate model.

Seasonal Differencing

The Christmas shopping season is accompanied by an unusually large number of transactions, and the Federal Reserve expands the money supply to accommodate the increased demand for money. As shown by the dashed line in Figure 2.7, the U.S. money supply, as measured by M1, has a decidedly upward trend. The series, called M1NSA, is contained in the file QUARTERLY.XLS. You can use the data to follow along with the discussion below. The logarithmic change, shown by the solid line, appears to be stationary. Nevertheless, there is a clear seasonal pattern in that the value of the fourth quarter for any year is substantially higher than that for the adjacent quarters.

This combination of strong seasonality and nonstationarity is often found in economic data. The ACF for a process with strong seasonality is similar to that for a nonseasonal process; the main difference is that the spikes at lags $s, 2s, 3s, \dots$, do not exhibit rapid decay. We know that it is necessary to difference (or take the logarithmic change of) a nonstationary process. Similarly, if the autocorrelations at the seasonal lags do not decay, it is necessary to take the seasonal difference so that the other autocorrelations are not dwarfed by the seasonal effects. The ACF and PACF for the growth rate of M1 are shown in Panel (a) of Figure 2.8. For now, just focus on the autocorrelations at the seasonal lags. All seasonal autocorrelations are large and show no tendency to decay. In particular, $\rho_4 = 0.58$, $\rho_8 = 0.50$, $\rho_{12} = 0.38$, $\rho_{16} = 0.34$, $\rho_{20} = 0.34$, and $\rho_{24} = 0.37$. As should be clear from the figure, these autocorrelations are larger than any of those at nonseasonal frequencies.

The first step in the Box–Jenkins method is to transform the data so as to make it stationary. As such, a logarithmic transformation is helpful because it can straighten

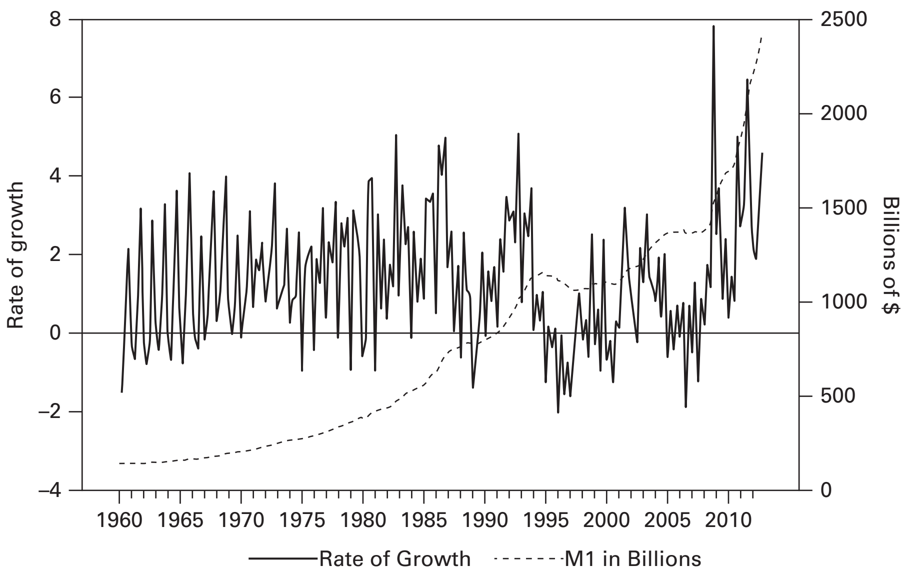


FIGURE 2.7 The Level and Growth Rate of M1

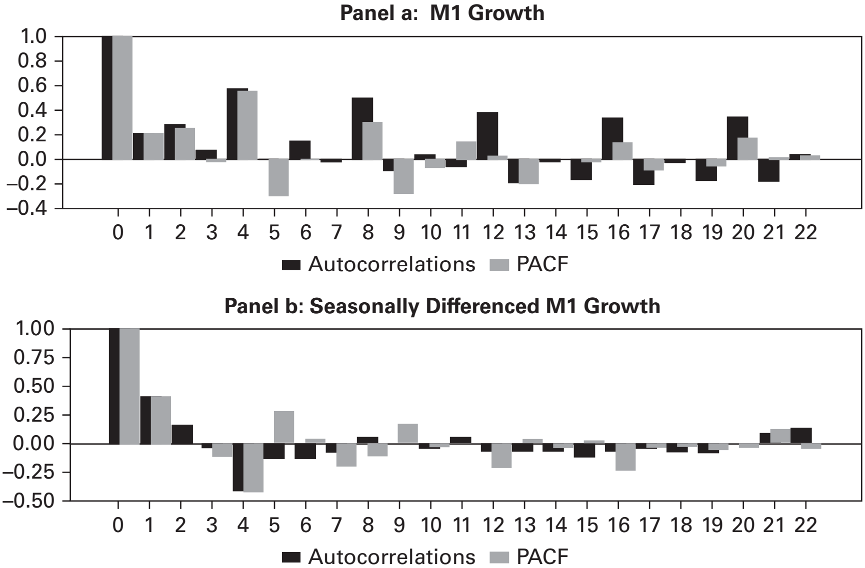


FIGURE 2.8 ACF and PACF

the nonlinear trend in M1. Let y_t denote the log of M1. As mentioned above, the first difference of the $\{y_t\}$ sequence, illustrated by the solid line in Figure 2.7, appears to be stationary. However, to remove the strong seasonal persistence in the data, we need to take the seasonal difference. For quarterly data, the seasonal difference is $y_t - y_{t-4}$. Since the order of differencing is irrelevant, we can form the transformed sequence

$$m_t = (1 - L)(1 - L^4)y_t$$

Thus, we use the seasonal difference of the first difference. The ACF and PACF for the $\{m_t\}$ sequence are shown in Panel (b) of Figure 2.8; the properties of this series are much more amenable to the Box–Jenkins methodology. The autocorrelation and partial autocorrelations for the first few lags are strongly suggestive of an AR(1) process ($\rho_1 = \phi_{11} = 0.41$, $\rho_2 = 0.16$, and $\phi_{22} = -0.01$). Recall that the ACF for an AR(1) process will decay and the PACF will cut to zero after lag 1. Given that $\rho_4 = -0.42$, $\rho_5 = -0.14$, $\phi_{44} = -0.44$, and $\phi_{55} = 0.28$, there is evidence of remaining seasonality in the $\{m_t\}$ sequence. The seasonal term is most likely to be in the form of an MA coefficient since the autocorrelation cuts to zero, whereas the PACF does not. Nevertheless, it is best to estimate several similar models and then select the best. Estimates of the following three models are reported in Table 2.5:

$m_t = a_0 + a_1 m_{t-1} + \varepsilon_t + \beta_4 \varepsilon_{t-4}$ Model 1: AR(1) with Seasonal MA

$m_t = a_0 + (1 + a_1 L)(1 + a_4 L^4) m_{t-1} + \varepsilon_t$ Model 2: Multiplicative Autoregressive

$m_t = a_0 + (1 + \beta_1 L)(1 + \beta_4 L^4)\varepsilon_t$ Model 3: Multiplicative Moving Average

The point estimates of the coefficients all imply stationarity and invertibility. Moreover, except for the intercepts, all are at least six standard deviations from zero. However, the diagnostic statistics all suggest that model 1 is preferred. Model 1 has the best

Table 2.5 Three Models of Money Growth

	Model 1	Model 2	Model 3
a_1	0.541	0.496	
	(8.59)	(7.66)	
a_4		−0.476	
		(−7.28)	
β_1			0.453
			(6.84)
β_4	−0.759		−0.751
	(−15.11)		(−14.87)
SSR	0.0177	0.0214	0.0193
AIC;	−735.9;	−701.3;	−720.1;
SBC	−726.2	−691.7	−710.4
$Q(4)$	1.39 (0.845)	3.97 (0.410)	22.19 (0.000)
$Q(8)$	6.34 (0.609)	24.21 (0.002)	30.41 (0.000)
$Q(12)$	14.34 (0.279)	32.75 (0.001)	42.55 (0.000)

To ensure comparability, the three models are estimated over the 1962*Q*3 – 2008*Q*2 period. The estimated intercepts are not reported since all were insignificantly different from zero. The figures in parentheses following the *Q*-statistics are significance levels.

fit in that it has the lowest SSR, AIC, and SBC. Moreover, the *Q*-statistics for lags 4, 8, and 12 indicate that the residual autocorrelations are insignificant. In contrast, the residual correlations for model 2 are significant at the long lags [i.e., $Q(8)$ and $Q(12)$ are significant at the 0.022 and 0.002 levels]. This is because the multiplicative seasonal autoregressive (SAR) term does not adequately capture the seasonal pattern. An SAR term implies autoregressive decay from period s into period $s+1$. In Panel (b) of Figure 2.8, the value of ρ_4 is −0.42 but ρ_5 is quite small. As such, a multiplicative seasonal moving-average (SMA) term might be more appropriate. Model 3 properly captures the seasonal pattern, but the MA(1) term does not capture the autoregressive decay present at the short lags. Other diagnostic methods, including splitting the sample, suggest that model 1 is appropriate.

The out-of-sample forecasts are shown in Figure 2.9. To create the one- through twelve-step-ahead forecasts, model 1 was estimated over the full sample period 1961*Q*3–2012*Q*4. The estimated model is

$$m_t = 0.545m_{t-1} + \varepsilon_t - 0.765\,\varepsilon_{t-4} \tag{2.69}$$

Given that $m_{2012:4} = -0.00176$ and the residual for 2012:1 was 0.00272 (i.e., $\hat{\varepsilon}_{2012:1} = 0.00272$, the forecast of $m_{2013:1}$ is −0.00304. Now, use this forecast and the value of $\hat{\varepsilon}_{2012:2}$ to forecast $m_{2013:2}$. You can continue in this fashion so as to obtain the out-of-sample forecasts for the $\{m_t\}$ sequence. Although you do not have the residuals for periods beyond 2012:4, you can simply use their forecasted values of zero. The trick to forecasting future values of M1 from the $\{m_t\}$ sequence is to sum the changes and the seasonal changes so as to obtain the logarithm of the forecasted values of M1. Since $m_t = (1-L)(1-L^4)\ln(\text{M1}_t)$, it follows that the value of $\ln(\text{M1}_t)$ can be

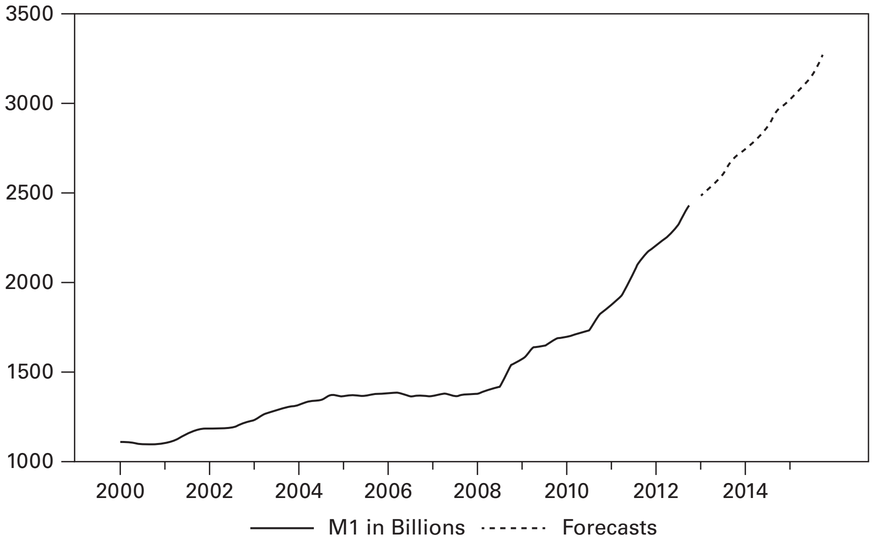


FIGURE 2.9 Forecasts of M1

obtained from $m_t + \ln(\text{M1}_{t-1}) + \ln(\text{M1}_{t-4}) - \ln(\text{M1}_{t-5})$. The first 12 of the forecasted values are plotted in Figure 2.9.

The procedures illustrated in this example with highly seasonal data are typical of many other series. With highly seasonal data, it is necessary to supplement the Box–Jenkins method:

1. In the identification stage, it is usually necessary to seasonally difference the data and to check the ACF of the resultant series. Often, the seasonally differenced data will not be stationary. In such instances, the data may also need to be first differenced.
2. Use the ACF and PACF to identify potential models. Try to estimate models with low-order nonseasonal ARMA coefficients. Consider both additive and multiplicative seasonality. Allow the appropriate form of seasonality to be determined by the various diagnostic statistics.

A compact notation has been developed that allows for the efficient representation of intricate models. As mentioned in the previous sections, the dth difference of a series is denoted by Δ^d. Hence,

$$\begin{aligned}\Delta^2 y_t &= \Delta(y_t - y_{t-1}) \\ &= y_t - 2y_{t-1} + y_{t-2}\end{aligned}$$

A seasonal difference is denoted by Δ_s where s is the period of the data. The Dth such seasonal difference is Δ_s^D. For example, if we want the second seasonal difference of a monthly series, we can form

$$\begin{aligned}\Delta_{12}^2 y_t &= \Delta_{12}(y_t - y_{t-12}) \\ &= \Delta_{12} y_t - \Delta_{12} y_{t-12}\end{aligned}$$

$$= y_t - y_{t-12} - (y_{t-12} - y_{t-24})$$
$$= y_t - 2y_{t-12} + y_{t-24}$$

Combining the two types of differencing yields $\Delta^d \Delta_s^D$. Multiplicative models are written in the form ARIMA$(p, d, q)(P, D, Q)_s$

where: p and q = the nonseasonal ARMA coefficients
d = number of nonseasonal differences
P = number of multiplicative autoregressive coefficients
D = number of seasonal differences
Q = number of multiplicative moving-average coefficients
s = seasonal period.

Using this notation, we can say that the fitted equation for $m_t = \Delta\Delta_4^1 \ln(\mathrm{M1}_t)$ is an ARIMA$(1, 1, 0)(0, 1, 1)_s$ model. In applied work, the ARIMA$(1, 1, 0)(0, 1, 1)_s$ and the ARIMA$(0, 1, 1)(0, 1, 1)_s$ models occurs routinely; the latter is called the *airline model* ever since Box and Jenkins (1976) used this model to analyze airline travel data.

12. PARAMETER INSTABILITY AND STRUCTURAL CHANGE

One key assumption of the Box–Jenkins methodology is that the structure of the data-generating process does not change. As such, the values of the a_i and β_i should be constant from one period to the next. However, in some circumstances, there may be reasons to suspect a structural break in the data-generating process. For example, in a model of GDP growth, it seems natural to inquire whether the oil price shocks of 1973, the events surrounding the tragedy of September 2011, and/or the financial crisis of 2008 had any significant impacts on the coefficients. Of course, parameter instability need not result from a single discrete event. The recent evidence concerning climate change suggests that weather sensitive series such as crop yields, rainfall, and the number of ski days at Snowmass are most likely to be affected in a sustained, but gradual, way.

Testing for Structural Change

If you have reason to suspect a structural break at a particular date, it is straightforward to use a Chow test. The essence of the Chow test is to fit the same ARMA model the prebreak data and to the postbreak data. If the two models are not sufficiently different, it can be concluded that there has not been any structural change in the data-generating process.

In general, suppose you estimated an ARMA(p, q) model using a sample size of T observations. Denote the sum of the squared residuals as SSR. Also, suppose that you have reason to suspect a structural break immediately following date t_m. You can perform a Chow test by dividing the T observations into two subsamples with t_m observations in the first and $t_n = T - t_m$ observations in the second. Use each subsample to estimate the two models:

$$y_t = a_0(1) + a_1(1)y_{t-1} + \cdots + a_p(1)y_{t-p} + \varepsilon_t + \beta_1(1)\varepsilon_{t-1} + \cdots + \beta_q(1)\varepsilon_{t-q}$$
$$\text{using } t_1, \ldots, t_m$$
$$y_t = a_0(2) + a_1(2)y_{t-1} + \cdots + a_p(2)y_{t-p} + \varepsilon_t + \beta_1(2)\varepsilon_{t-1} + \cdots + \beta_q(2)\varepsilon_{t-q}$$
$$\text{using } t_{m+1}, \ldots, t_T$$

Let the sum of the squared residuals from each model be SSR_1 and SSR_2, respectively. To test the restriction that all coefficients are equal [i.e., $a_0(1) = a_0(2)$ and $a_1(1) = a_1(2)$ and $\cdots a_p(1) = a_p(2)$ and $\beta_1(1) = \beta_1(2)$ and $\cdots \beta_q(1) = \beta_q(2)$], use an F-test and form:[6]

$$F = \frac{(\text{SSR} - \text{SSR}_1 - \text{SSR}_2)/n}{(\text{SSR}_1 + \text{SSR}_2)/(T - 2n)} \tag{2.70}$$

where n = number of parameters estimated ($n = p + q + 1$ if an intercept is included and $p + q$ otherwise) and the number of degrees of freedom are $(n, T - 2n)$.

Intuitively, if the restriction is not binding (i.e., if the coefficients are equal), the sum $\text{SSR}_1 + \text{SSR}_2$ should equal the sum of the squared residuals from the entire sample estimation. Hence, F should equal zero. The larger the calculated value of F, the more restrictive is the assumption that the coefficients are equal.

Of course, the method requires that there be a reasonable number of observations in each subsample. If either t_m or t_n is very small, the estimated coefficients will have little precision. An alternative type of Chow test is to a use dummy variable to detect a break in one of more of the coefficients. For example, if a break is suspected right after period t_m, you can create a dummy variable, D_t, such that $D_t = 0$ for all $t \le t_m$ and $D_t = 1$ for $t > t_m$. To test for a break in the intercept of an AR(1) model, for example, check for the significance of D_t in the regression $y_t = a_0 + \alpha_0 D_t + a_1 y_{t-1} + \varepsilon_t$. To allow for a break in both coefficients, also create the variable $D_t y_{t-1}$ and estimate the regression equation $y_t = a_0 + \alpha_0 D_t + a_1 y_{t-1} + \alpha_1 D_t y_{t-1} + \varepsilon_t$. You can test for a break by examining the individual t-statistics of α_0 and α_1 and the F-statistic for the null hypothesis $\alpha_0 = \alpha_1 = 0$.

Return to the example of the interest rate spread examined in Section 10. Suppose that there is reason to believe a break occurred at the end of 1981Q4. Consider the estimates for the two subperiods:

$$s_t = 0.923 + 0.367s_{t-1} + 0.285s_{t-2} + \varepsilon_t + 0.815\varepsilon_{t-1} - 0.153\varepsilon_{t-7}$$
$$(1960Q3\text{–}1981Q4)$$

and

$$s_t = 1.799 + 0.800s_{t-1} + 0.053s_{t-2} + \varepsilon_t + 0.354\varepsilon_{t-1} + 0.097\varepsilon_{t-7}$$
$$(1982Q1\text{–}2008Q1)$$

Although the coefficients of the models appear to be dissimilar, we can formally test for the equality of coefficients using (2.70). Respectively, the sum of squared residuals for the two equations are $\text{SSR}_1 = 27.564$ and $\text{SSR}_2 = 21.414$. Estimating the model over the full sample period yields $\text{SSR} = 49.692$. Since there are 191 usable observations in the sample and $n = 5$ (the intercept plus the four estimated coefficients), (2.70) becomes

$$F = [(49.692 - 27.564 - 21.414)/5]/[(27.564 + 21.414)/(191 - 10)] = 0.527.$$

With 5 degrees of freedom in the numerator and 181 in the denominator, we cannot reject the null of no structural change in the coefficients (i.e., we can accept the hypothesis that there is no structural change in the coefficients).

Alternatively, to test for a break in the intercept only, we can create the dummy variable D_t equal to zero prior to 1982Q1 and equal to unity beginning in 1982Q1. Now, consider the equation for the spread estimated over the entire 1960Q1–2008Q1 period

$$\begin{array}{llllllll} s_t = & 1.277 & + 0.312D_t & + 0.336s_{t-1} & + 0.435s_{t-2} & + \varepsilon_t & + 0.837\varepsilon_{t-1} & - 0.134\varepsilon_{t-7} \\ & (3.55) & (0.82) & (3.23) & (4.43) & & (13.14) & (-3.33) \end{array}$$

Since D_t jumps from 0 to 1 in 1982Q1, the estimate for the intercept is 1.227 prior to 1982Q1 and 1.589 (= 1.277 + 0.312) beginning in 1982Q1. However, since the t-statistic for the null hypothesis $D_t = 0$ cannot be rejected, there is no evidence of a significant intercept break.

Endogenous Breaks

The Chow test asks whether there is a break beginning at some particular known break date t_m. A break occurring at a date not prespecified by the researcher is called an **endogenous break** to denote that the fact that it was not the result of a fixed break date such as September 2011. To determine whether there is a break anywhere in the sample, you could perform a Chow test for every potential break date t_m. It should not be surprising that the break date that results in the largest value of the F-statistic provides a consistent estimate of the actual break date, if any. In order to ensure an adequate number of observations in each of the two subsamples, it is necessary to have a "trimming" such that the break could not occur before the first t_0 observations or after the last $T - t_0$ observations. In applied research, it is common to use a trimming value of 10% so that there are at least 10% of the observations in each of the two subsamples. In the interest rate spread example, there are 191 usable observations in the 1960Q1–2008Q1 period (since the first two are lost when estimating the coefficient for s_{t-2}). If you used a 10% trimming, you could check for a break everywhere in the interval 1965Q1–2003Q2 (each about 19 observations from the beginning and end of the usable data). Unfortunately, searching for the most likely break date means that the F-statistic for the null hypothesis of no break is inflated. After all, you have just searched for the date that leads to the maximum, or supremum, value of the sample F-statistic. As such, the distribution for the F-statistic is not standard and cannot be obtained from a traditional F-table. As detailed in Chapter 7, a number of papers, including Andrews and Ploberger (1994) and Hansen (1997), show how to obtain the appropriate critical values. Fortunately, a number of software packages can readily perform such tests.

Parameter Instability

Notice that the Chow test and its variants require the researcher to specify a particular break date and to assume that the break fully manifests itself at that date. The intercept, for example, is $a_0(1)$ up to t_m and is precisely $a_0(2)$ beginning at t_{m+1}. However, the assumption that a break occurs exactly at a single point in time may not always be appropriate. As mentioned above, there is no particular date at which we can say that significant climate change has occurred. Similarly, it is not clear how we can provide a specific break date to denote the advent "financial deregulation" in the asset markets or to assign a specific date to the development of the microcomputer. These are processes that have been evolving over time. Even if we could date the precise start of financial deregulation or the computer revolution, the full effects of these changes would not occur instantly. As such, it should not be surprising that a number of procedures have been developed that check for parameter stability without the need to identify a particular break date. Probably, the simplest method is to estimate the model recursively. For example, if you have 150 observations, you can estimate the model using only the first few, say 10, observations. Plot the individual coefficients and then reestimate the model using the first 11 observations. You can keep repeating this process until you use all 150 observations. In general, the plots of the coefficients will not be flat since the preliminary values are estimated using a very small number of observations. However, after a "burn-in" period, the time plots of the individual coefficients can provide evidence of coefficient stability. If the magnitude of a coefficient suddenly begins to change, you should suspect a structural change at that point. A sustained change in a coefficient might indicate a model misspecification. One particularly helpful modification of this procedure is to plot each coefficient along with its estimated ± 2 standard deviation band. The bands represent confidence intervals for the estimated coefficients. In this way, it can be seen if the coefficients are always statistically significant and whether the coefficients in the early periods appear to be statistically different from those of the latter periods.

At each step along the way, it is also possible to create the one-step-ahead forecast error. Let $e_t(1)$ be the one-step-ahead forecast error made using all observations through t. In other words, $e_t(1)$ is the difference between y_{t+1} and your conditional forecast of y_{t+1} (i.e., $E_t y_{t+1}$). If you start with the first 10 observations, the value of $e_{10}(1)$ will be $y_{11} - E_{10}y_{11}$ and the value of $e_{149}(1)$ will be $y_{150} - E_{149}y_{150}$. [Note: If you understand the notation, it should be clear that you cannot create the value $e_{150}(1)$ since you do not have the value of y_{151}.] If your model fits the data well, the forecasts should be unbiased so that the sum of these forecast errors should not be "too far" from zero. In fact, Brown, Durbin, and Evans (1975) calculate whether the cumulated sum of the forecast errors is statistically different from zero. To be a bit more formal, define:

$$\text{CUSUM}_N = \sum_{i=n}^{N} e_i(1)/\sigma_e \qquad N = n, \ldots, T-1$$

where n denotes the date of the first forecast error you constructed, T denotes the date of the last observation in the data set, and σ_e is the estimated standard deviation of the forecast errors. With 150 total observations ($T = 150$), if you start the procedure using the first 10 observations ($n = 10$), 140 forecast errors ($T - n$) can be created. Note that σ_e is created using all $T - n$ forecast errors. If $n = 10$, to create CUSUM_{10}, use the first 10 observations to create the one-step-ahead forecast error and construct $e_{10}(1)/\sigma_e$. Now let $N = 11$ and create CUSUM_{11} as $[e_{10}(1) + e_{11}(1)]/\sigma_e$. Similarly, $\text{CUSUM}_{T-1} = [e_{10}(1) + \cdots + e_{T-1}(1)]/\sigma_e$. If you use the 5% significance level, the plot value of each value of CUSUM_N should be within a band of approximately $\pm\, 0.948\, [(T - n)^{0.5} + 2(N - n)(T - n)^{-0.5}]$.

An Example of a Break

In order to illustrate a breaking series, the first panel of Figure 2.10 shows 150 observations of the simulated series $y_t = 1 + 0.5y_{t-1} + \varepsilon_t$ for $t < 101$ and $y_t = 2.5 + 0.65y_{t-1} + \varepsilon_t$ for $t \geq 101$. The series is contained in the file Y_BREAK.XLS. Of course, in applied work, the break may not be so readily apparent. If you ignore the break and estimate the entire series as an AR(1) process, you should obtain

$$\begin{aligned} y_t = &\ 0.4442 + 0.8822y_{t-1} \\ &(2.635) \quad (22.764) \end{aligned}$$

As indicated by the remaining two panels of the figure, the estimated AR(1) model is seriously misspecified. Panel 2 shows the estimates of the AR(1) coefficient (along with their ± 2 standard deviation bands) resulting from a recursive estimation. The initial confidence intervals are quite wide since the first few estimations use a very small number of observations. The estimates all seem reasonable until about $t = 100$. At this point, the estimates of the AR(1) coefficient rise (the reverse of what happens in the data-generating process). Note that the confidence bands do not even overlap those from the middle periods. The clear suggestion is that there has been a significant structural change. The CUSUMs, shown in Panel (3), are clearly within the 90% confidence interval for $t < 101$. At this point, they begin to drift upward and depart from the band at $t = 125$. As such, the hypothesis of coefficient stability can be rejected.

Notice that the CUSUMs do not actually depart from the band until late in the sample. This is indicative of the problem that the CUSUM test may not detect coefficient instability occurring late in the sample period. Moreover, the test may not have much power if there are multiple changes with little overall effect on the CUSUMs. Nevertheless, the test is a useful diagnostic tool that does not require the researcher to stipulate the nature of the model's misspecification. It is able to detect model misspecifications from such varied sources including smooth structural breaks, multiple breaks, neglected nonlinearities in the data-generating process, or an overly parsimonious model. A variant of the test, often called CUSUM(2), is to form the CUSUMs using the squared errors. The use of the squared errors can help detect changes in the variance.

If you had strong reason to believe that the break occurred in period 101, you could form a dummy variable $D_t = 0$ from $t = 1$ to 100 and $D_t = 1$ thereafter. To check for

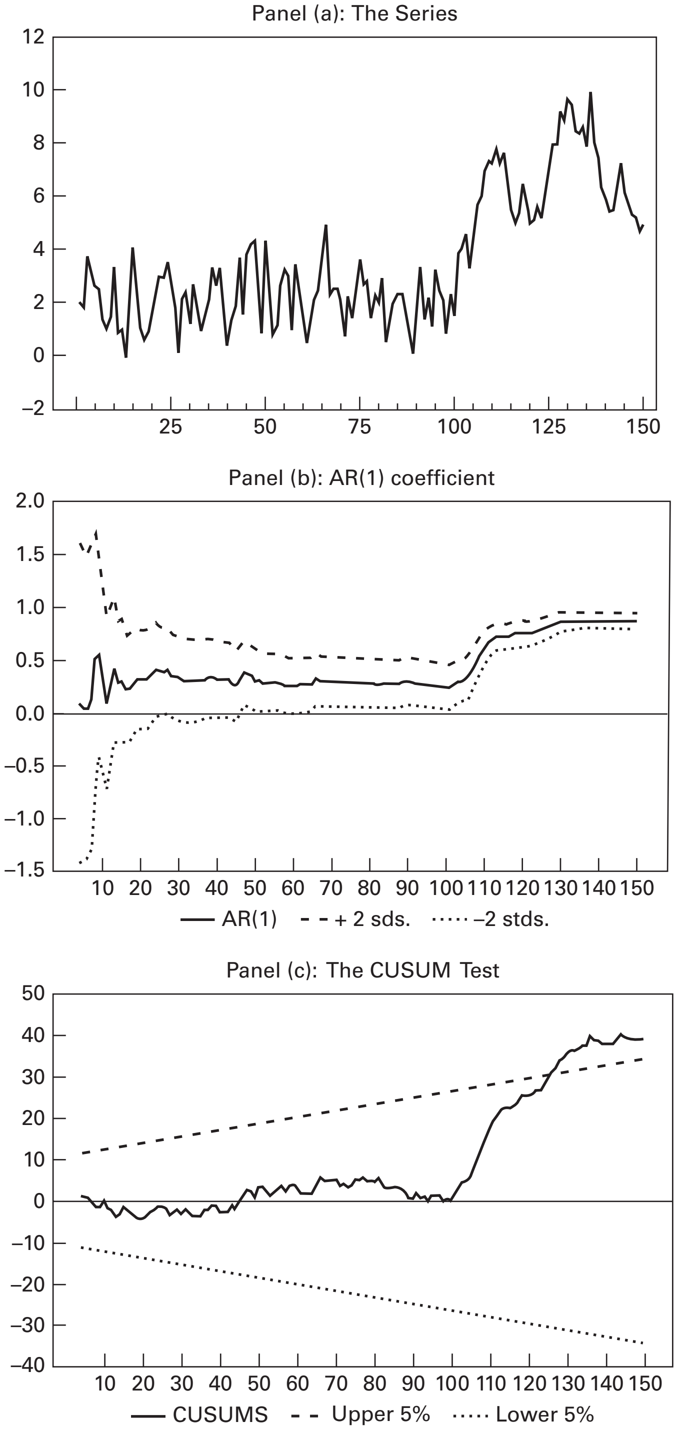


FIGURE 2.10 Recursive Estimation of the Model

an intercept break, estimate

$$\underset{}{y_t} = \underset{(5.36)}{0.9254} + \underset{(8.91)}{0.5683y_{t-1}} + \underset{(5.88)}{1.936D_t}$$

Since the coefficient for D_t is highly significant, you can conclude that there was a break in the intercept. To check for a break in the intercept and slope coefficient, also form the variable $D_t y_{t-1}$ and estimate:

$$y_t = \underset{(7.22)}{1.6015} + \underset{(2.76)}{0.2545y_{t-1}} - \underset{(-0.391)}{0.2244D_t} + \underset{(4.47)}{0.5433D_t y_{t-1}}$$

In this particular case, the dummy variables indicate that there is a break but do not measure the size of the break very well (Note: The actual break in the intercept is +1.5 and the actual break in the AR(1) coefficient is 0.15.) The coefficient for the intercept break is not significant while the break in the slope coefficient is highly significant. The F-statistic for the joint hypothesis that the coefficients on D_t and $D_t y_{t-1}$ are equal to zero is 29.568. With 2 degrees of freedom in the numerator and 145 in the denominator, this value is significant at any conventional level. The important point is that you can conclude that the simple AR(1) model is misspecified because of a structural break.

If you wanted to estimate the most likely value for t_m, you could repeat the estimation for every time period in the interval $15 < t_m < 135$. The values of the F-statistics from each recursive estimation are shown in Figure 2.11. Notice that the F-values are largest for $t_m = 100$. Although this consistent estimate of the break date turns out to be exactly correct, you should expect a discrepancy when using actual data. Also note that the F-test (and the t-statistics of the individual coefficients) for the null hypothesis of no structural change can be tested using Hansen's (1997) method (see Chapter 7).

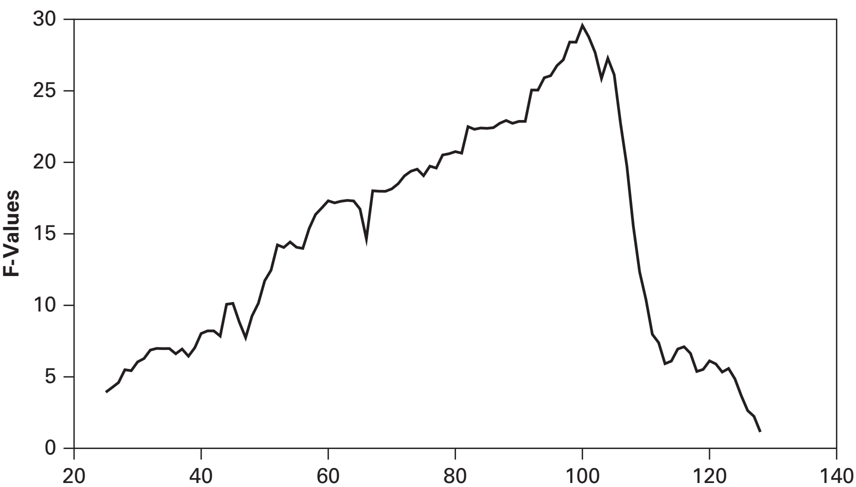


FIGURE 2.11 Recursive F-tests

13. COMBINING FORECASTS

What should you do if you have several plausible models and want to use them to forecast? For example, in Section 10, it turned out that there are several plausible models of the interest rate spread. It makes little sense to forecast only with the "best" model and discard the others. After all, the other models may capture some information that is not contained in the others. The natural answer is to forecast with all of the plausible models and then take the average of the forecasts.

In turns out that the intuitive notion of using the average, or composite, forecast is quite reasonable. Bates and Granger (1969) were among the first to confirm the intuition that a weighted average of forecasts can be quite beneficial. Let the series f_{it} contain the one-step-ahead forecasts of y_t from model i ($i = 1, 2, \dots, n$). Consider the composite forecast f_{ct} constructed as weighted average of the individual forecasts

$$f_{ct} = w_1 f_{1t} + w_2 f_{2t} + \cdots + w_n f_{nt} \tag{2.71}$$

where w_i are weights such that $\sum_{i=1}^{n} w_i = 1$.

If the forecasts are unbiased (so that $E_{t-1} f_{it} = y_t$), it follows that the composite forecast is also unbiased:

$$\begin{aligned} E_{t-1} f_{ct} &= w_1 E_{t-1} f_{1t} + w_2 E_{t-1} f_{2t} + \cdots + w_n E_{t-1} f_{nt} \\ &= w_1 y_t + w_2 y_t + \cdots + w_n y_t = y_t \end{aligned}$$

To keep the notation simple, return to the case in which $n = 2$. Subtract y_t from each side of (2.71) to obtain

$$f_{ct} - y_t = w_1(f_{1t} - y_t) + (1 - w_1)(f_{2t} - y_t)$$

Now let e_{1t} and e_{2t} denote the series containing the one-step-ahead forecast errors from models 1 and 2 (i.e., $e_{it} = y_t - f_{it}$) and let e_{ct} be the composite forecast error. As such, we can write

$$e_{ct} = w_1 e_{1t} + (1 - w_1) e_{2t}$$

The variance of the composite forecast error is

$$\mathrm{var}(e_{ct}) = w_1^2\, \mathrm{var}(e_{1t}) + (1 - w_1)^2 \mathrm{var}(e_{2t}) + 2w_1(1 - w_1)\mathrm{cov}(e_{1t}e_{2t}) \tag{2.72}$$

At this point, you should be able to see the potential benefits of combining such forecasts. To take a simple example, suppose that the forecast error variances are the same size and that $\mathrm{cov}(e_{1t}e_{2t}) = 0$. If you take a simple average by setting $w_1 = 0.5$, (2.72) indicates that the variance of the composite forecast is 25% of the variances of either forecast: $\mathrm{var}(e_{ct}) = 0.25\mathrm{var}(e_{1t}) = 0.25\mathrm{var}(e_{2t})$.

Optimal Weights

Although simple averaging can work to reduce the forecast error variance, it is possible to find the optimal weights. If we use (2.72) and select the weight w_1 so as to minimize

$\text{var}(e_{ct})$:

$$\frac{\delta \text{ var}(e_{ct})}{\delta w_1} = 2w_1 \text{ var}(e_{1t}) - 2(1 - w_1) \text{ var}(e_{2t}) + 2(1 - 2w_1) \text{ cov}(e_{1t}e_{2t})$$

The optimal value of w_1 (called w_1^*) is

$$w_1^* = \frac{\text{var}(e_{2t}) - \text{cov}(e_{1t}e_{2t})}{\text{var}(e_{1t}) + \text{var}(e_{2t}) - 2 \text{ cov}(e_{1t}e_{2t})} \tag{2.73}$$

and if $\text{cov}(e_{1t}e_{2t}) = 0$, w_1^* can be written as

$$w_1^* = \frac{\text{var}(e_{2t})}{\text{var}(e_{1t}) + \text{var}(e_{2t})} = \frac{\text{var}(e_{1t})^{-1}}{\text{var}(e_{1t})^{-1} + \text{var}(e_{2t})^{-1}}$$

Hence, if the covariance is zero, the optimal weight is inversely proportional to its variance. As $\text{var}(e_{1t})$ gets relatively small, the weight attached to f_{1t} goes to unity, and as $\text{var}(e_{1t})$ gets relatively large, the weight attached to f_{1t} goes to zero. Since the actual forecast error variances are not known, in practice, they replaced by the estimated forecast error variances from the type of out-of-sample forecast exercises conducted above. In a setting with a large number of competing forecasting models, constructing optimal weights as in (2.73) can be quite tedious. Moreover, estimates of the covariance terms are often poor. As such, a number of researchers, including Bates and Granger (1969), recommend constructing the weights excluding the covariance terms. Hence, in the n-variable case, the weights can be constructed as

$$w_n^* = \frac{\text{var}(e_{1t})^{-1}}{\text{var}(e_{1t})^{-1} + \text{var}(e_{2t})^{-1} + \cdots + \text{var}(e_{nt})^{-1}} \tag{2.74}$$

It is straightforward to compute the reciprocals of forecast error variance from each model and then to normalize each by the sum across all of the models. Granger and Ramanathan (1984) show that an equivalent method for constructing the weights is to use a regression model. Consider the regression equation

$$y_t = \alpha_0 + \alpha_1 f_{1t} + \alpha_2 f_{2t} + \cdots + \alpha_n f_{nt} + v_t \tag{2.75}$$

Of course, it would be possible to force $\alpha_0 = 0$ and $\alpha_1 + \alpha_2 + \cdots + \alpha_n = 1$. Under these conditions, the α_i's would have the direct interpretation of optimal weights so that w_i^* could be set equal to α_i. However, Granger and Ramanathan recommend the inclusion of an intercept to account for any bias and to leave the α_i's unconstrained. As surveyed in Clemen (1989), not all researchers agree with the Granger–Ramanathan recommendation and a substantial amount of work has been conducted so as to obtain optimal weights.

There are two important differences between (2.74) and (2.75). In (2.75), one or more of the estimated weights may be negative. In such circumstances, most researchers would reestimate the regression without the forecast associated with the most negative coefficient. Moreover, in (2.75), the $\{v_t\}$ sequence can be serially correlated. In such circumstances, Diebold (1988) recommends using lagged values of y_t and/or moving average terms to capture the serial correlation.

It is also possible to use the SBC as a weighting factor. Now the weights are determined by in-sample fit instead of out-of-sample forecasts. Let SBC_i be the SBC from model i and let SBC^* be the SBC from the best fitting model. You can easily form $\alpha_i = \exp[(\text{SBC}^* - \text{SBC}_i)/2]$ and then construct the weights

$$w_i^* = \alpha_i / \sum_{t=1}^{n} \alpha_i$$

Since exp(0) = 1, the model with the best fit has the weight $1/\Sigma\alpha_i$. Since α_i is decreasing in the value of SBC_i, models with a poor fit have smaller weights than models with large values of the SBC.

Example Using the Spread

In Section 10, we examined seven different ARMA models of the interest rate spread. Given that the data ends in April 2012, if you were to use each of the seven models to make a one-step-ahead forecast for January 2013, you should find

	AR(7)	AR(6)	AR(2)	AR(‖1,2,7‖)	ARMA(1, 1)	ARMA(2, 1)	ARMA(2,‖1,7‖)
$f_{i2013:1}$	0.775	0.775	0.709	0.687	0.729	0.725	0.799

Simple averaging of the individual forecasts (i.e., setting all weights equal to 1/7) results in a combined forecast of 0.743. Now, use the methodology discussed in Section 10 to construct 50 one-step-ahead out-of-sample forecasts for each of the seven models so as to obtain the f_{it} series. After constructing the e_{it} sequences as $f_{it} - y_t$, it is trivial to find the seven values of $\text{var}(e_{it})$. If you use (2.74), you should find that the forecast error variances and the associated weights are

	AR(7)	AR(6)	AR(2)	AR(‖1,2,7‖)	ARMA(1, 1)	ARMA(2, 1)	ARMA(2,‖1,7‖)
$\text{var}(e_{it})$	0.635	0.618	0.583	0.587	0.582	0.600	0.606
w_i	0.135	0.139	0.147	0.146	0.148	0.143	0.141

The weights are very similar because the forecast error variances are alike. Weighting the individual forecasts yields the composite forecast $f_{c2013:1} = 0.741$.

Next, use the spread (s_t) to estimate a regression in the form of (2.75). If you omit the intercept and constrain the weights to unity, you should obtain

$$s_t = 0.55f_{1t} - 0.25f_{2t} - 2.37f_{3t} + 2.44f_{4t} + 0.84f_{5t} - 0.28f_{6t} + 1.17f_{7t} \qquad (2.76)$$

Although some researchers would include the negative weights in (2.76), most would eliminate those that are negative. If you successively reestimate the model by eliminating the forecast with the most negative coefficient, you should obtain

$$s_t = 0.326f_{4t} + 0.170f_{5t} + 0.504f_{7t}$$

All of the coefficients are positive and the residuals do not show any sign of serial correlation. As such, it is reasonable to use the weights 0.326, 0.170, and 0.504 for the forecasts from the AR(‖1, 2, 7‖), ARMA(1, 1), and ARMA(2, ‖1, 7‖), models, respectively. The composite forecast using the regression method is 0.326(0.687) + 0.170(0.729) + 0.504(0.799) = 0.751.

Finally, if you use the values of the SBC as weights, you should obtain

	AR(7)	AR(6)	AR(2)	AR(||1,2,7||)	ARMA(1, 1)	ARMA(2, 1)	ARMA(2,||1,7||)
w_i	0.000	0.000	0.011	0.001	0.112	0.103	0.773

The composite forecast using SBC weights is 0.782. In actuality, the spread in 2013:1 turned out to be 0.74 (the actual data contains only two decimal places). Of the four methods, simple averaging and weighting by the forecast error variances did quite well. In this instance, the regression method and constructing the weights using the SBC provided the worst composite forecasts.

14. SUMMARY AND CONCLUSIONS

The chapter focuses on the Box–Jenkins (1976) approach to identification, estimation, diagnostic checking, and forecasting a univariate time series. ARMA models can be viewed as a special class of linear stochastic difference equations. By definition, an ARMA model is covariance stationary in that it has a finite and time-invariant mean and covariances. For an ARMA model to be stationary, the characteristic roots of the difference equation must lie inside the unit circle. Moreover, the process must have started infinitely far in the past or the process must always be in equilibrium.

In the identification stage, the series is plotted, and the sample autocorrelations and partial correlations are examined. A slowly decaying autocorrelation function suggests nonstationarity behavior. In such circumstances, Box and Jenkins recommend differencing the data. Formal tests for nonstationarity are presented in Chapter 4. A common practice is to use a logarithmic or Box–Cox transformation if the variance does not appear to be constant. Chapter 3 presents some modern techniques that can be used to model the variance.

The sample autocorrelations and partial correlations of the suitably transformed data are compared to those of various theoretical ARMA processes. All plausible models are estimated and compared using a battery of diagnostic criteria. A well-estimated model (i) is parsimonious, (ii) has coefficients that imply stationarity and invertibility, (iii) fits the data well, (iv) has residuals that approximate a white-noise process, (v) has coefficients that do not change over the sample period, and (vi) has good out-of-sample forecasts.

A useful check for coefficient instability involves recursive estimation techniques. A sudden change in the recursive estimates of one or more coefficients is indicative of a structural break. The Chow test can be used to test for a break at a known date and more gradual changes can be detected by recursive estimation or by a CUSUM test. As discussed in Chapter 7, the Andrews and Ploberger (1994) test can detect an endogenous break. Bai and Perron (1998, 2003) show how to test for multiple endogenous breaks.

In utilizing the Box–Jenkins methodology, you will find yourself making many seemingly *ad hoc* choices. The most parsimonious model may not have the best fit but may have the best out-of-sample forecasts. You will find yourself addressing the following types of questions: What is the most appropriate data transformation? Is an ARMA(2, 1) model more appropriate than an ARMA(1, 2) specification? How can seasonality best be modeled? What should be done about seemingly significant coefficients at reasonably long lags? Given this latitude, many view the Box–Jenkins methodology as an art rather than a science. Nevertheless, the technique is best learned through experience. The exercises at the end of this chapter are designed to guide you through the types of choices you will encounter in your own research.

QUESTIONS AND EXERCISES

1. In the coin-tossing example of Section 1, your average winnings on the last four tosses (w_t) can be denoted by

$$w_t = 1/4\varepsilon_t + 1/4\,\varepsilon_{t-1} + 1/4\,\varepsilon_{t-2} + 1/4\,\varepsilon_{t-3}$$

a. Find the expected value of your winnings. Find the expected value given that $\varepsilon_{t-3} = \varepsilon_{t-2} = 1$.
b. Find var(w_t). Find var(w_t) conditional on $\varepsilon_{t-3} = \varepsilon_{t-2} = 1$.
c. Find cov(w_t, w_{t-1}), cov(w_t, w_{t-2}), and cov(w_t, w_{t-5}).

2. Consider the second-order autoregressive process $y_t = a_0 + a_2 y_{t-2} + \varepsilon_t$ where $|a_2| < 1$.

a. Find:
i. $E_{t-2}y_t$ **ii.** $E_{t-1}y_t$ **iii.** $E_t y_{t+2}$ **iv.** $\text{cov}(y_t, y_{t-1})$
v. $\text{cov}(y_t, y_{t-2})$ **vi.** the partial autocorrelations ϕ_{11} and ϕ_{22}.
b. Find the impulse response function. Given y_{t-2}, trace out the effects of an ε_t shock on the $\{y_t\}$ sequence.
c. Determine the forecast function: $E_t y_{t+s}$. The forecast error $e_t(s)$ is the difference between y_{t+s} and $E_t y_{t+s}$. Derive the correlogram of the $\{e_t(s)\}$ sequence. {*Hint*: Find $E_t e_t(s)$, var[$e_t(s)$], and $E_t[e_t(s)e_t(s-j)]$ for $j = 0$ to s}.

3. Substitute (2.10) into $y_t = a_0 + a_1 y_{t-1} + \varepsilon_t$. Show that the resulting equation is an identity.

a. Find the homogeneous solution to $y_t = a_0 + a_1 y_{t-1} + \varepsilon_t$.
b. Find the particular solution given that $|a_1| < 1$.
c. Show how to obtain (2.10) by combining the homogeneous and particular solutions.

4. The general solution to an nth-order difference equation requires n arbitrary constants. Consider the second-order equation $y_t = a_0 + 0.75y_{t-1} - 0.125y_{t-2} + \varepsilon_t$.

a. Find the homogeneous and particular solutions. Discuss the shape of the impulse response function.
b. Find the values of the initial conditions that ensure the $\{y_t\}$ sequence is stationary.
c. Given your answer to part (b), derive the correlogram for the $\{y_t\}$ sequence.

5. Consider the second-order stochastic difference equation: $y_t = 1.5y_{t-1} - 0.5y_{t-2} + \varepsilon_t$.

a. Find the characteristic roots of the homogeneous equation.
b. Demonstrate that the roots of $1 - 1.5L + 0.5L^2$ are the reciprocals of your answer in part a.
c. Given initial conditions for y_0 and y_1, find the solution for y_t in terms of the current and past values of the $\{\varepsilon_t\}$ sequence.
d. Find the forecast function for y_{T+s} (i.e., find the solution for the values of y_{T+s} given the values of y_T and y_{T-1}).
e. Find Ey_t, Ey_{t+1}, var(y_t), var(y_{t+1}), and cov(y_{t+1}, y_t).